DSR-GCN: Differentiated-Scale Restricted Graph Convolutional Network for Few-Shot Hyperspectral Image Classification

Zhaohui Xue, *Member, IEEE*, Zhiwei Liu, and Mengxue Zhang

***Abstract*— Graph convolutional networks (GCNs) have shown great potential for few-shot hyperspectral image (HSI) classification. Mainstream GCNs construct graphs according to single-scale segmentation, which usually ignores subtle adjacency relations between small regions, leading to an unreliable initial local graph. To overcome the above issue, we propose a differentiated-scale restricted GCN (DSR-GCN) for HSI classification. First, we propose a differentiated-scale graph construction method considering both the subtle and relative wider range spectral–spatial relation. Second, restricted fusion loss is designed to restrict the fusion of features extracted with differentiated-scale GCN branches. Finally, we design a lightweight spectral-spatial siamese network (S3Net) to remedy local pixel-level features. The proposed DSR-GCN can better model spatial structure with a reliable and refined graph, and it can capture more discriminate features in few-shot learning (FSL) scenarios. Extensive experiments conducted on four benchmark datasets demonstrate that DSR-GCN outperforms the other deep learning methods in terms of classification accuracy and generalization performance, with improvements in terms of overall accuracy (OA) around 6.20%∼23.41% (Indian Pines), 4.45%∼16.48% (University of Pavia), 4.25%∼11.85% (Salinas), and 2.0%∼17.23% (University of Houston) under five labeled samples per class.**

***Index Terms*— Few-shot learning (FSL), graph convolutional network (GCN), hyperspectral image (HSI) classification, siamese network.**

I. Introduction

HYPERSPECTRAL image (HSI) contains hundreds of bands captured through the specific electromagnetic spectrum. With a very high spectral resolution, HSI can accurately distinguish similar materials in land cover at the pixel level [1]. Therefore, HSI classification has been widely applied to mineral identification, precision agriculture, military investigation, environmental monitoring, disaster preventing, and so on [2]. Traditional models, including support vector machine (SVM) [3], random forest (RF) [4], representation learning [5], and multiple kernel learning [6], focus on the handcrafted features.

Deep learning (DL) has the dominant ability to capture robust features in an end-to-end manner [7], [8], and typical DL models, including stacked autoencoder (SAE) [9], convolutional neural network (CNN) [10], deep belief network (DBN) [11], recurrent neural network (RNN) [12], generative adversarial networks (GANs) [13], capsule network [14], [15], and long short-term memory (LSTM) [16], have been rapidly introduced into the HSI classification task. Especially, CNN has gradually become an essential framework with various variants from 1-D CNN to 3-D CNN. In the early studies, 1-D CNN is used to extract vectorized spectral features with 1-D convolution kernel, which causes destruction of the original spectral dimension and naturally ignores the characteristics of the spatial distribution of HSI [17]. Therefore, 2-D CNN is later designed to explore the spatial information with 1-D convolution kernel and pooling layers [18]. Nevertheless, the spatial features learned by 2-D CNN are insufficient. Then, 3-D CNN is developed to simultaneously take spatial and spectral information into consideration with 3-D convolutions [19], [20], and the afterward methods, such as spectral–spatial unified network (SSUN) [21] and spectral–spatial residual network (SSRN) [22], follow and modify this line of thinking.

With the of CNNs, research on the improvement of the internal structure of CNN has gradually emerged. In terms of convolution kernel, deformable convolution [23], [24], multiscale convolution [25], [26], and dilated convolution [27] are explored to increase the receptive field of the fixed square convolution kernel. In terms of the regularization mechanism, nonlocal is introduced to capture the features of long-distance dependence [28]. In terms of the optimization method, the proxy-based framework is introduced to enhance the training speed and robustness of the DL model [29]. In addition, the attention mechanism has also made some progress. Among them, spatial attention mechanisms include self-attention [30], [31], multihead attention [32], [33], and nonlocal attention [34]. Spectral attention mechanism includes squeeze-and-excitation module [67]. Meanwhile, spatial and spectral attention methods have also been exploited for HSI classification [35], [36], [37].

Manuscript received 20 August 2022; revised 26 December 2022; accepted 27 February 2023. Date of publication 6 March 2023; date of current version 15 March 2023. This work was supported in part by the National Natural Science Foundation of China under Grant 41971279 and Grant 42271324, and in part by the Natural Science Foundation of Jiangsu Province under Grant BK20221506. *(Zhaohui Xue and Zhiwei Liu are co-first authors.) (Corresponding author: Zhaohui Xue.)*

Zhaohui Xue and Zhiwei Liu are with the School of Earth Sciences and Engineering, Hohai University, Nanjing 211100, China, and also with the Jiangsu Province Engineering Research Center of Water Resources and Environment Assessment Using Remote Sensing, Hohai University, Nanjing 211100, China (e-mail: zhaohui.xue@hhu.edu.cn).

Mengxue Zhang is with the Image and Signal Processing (ISP) Group, University of València, 46980 València, Spain.

Digital Object Identifier 10.1109/TGRS.2023.3253248

1558-0644 © 2023 IEEE. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

Although DL methods have shown great superiority for HSI classification, they are faced with the overfitting issue of learning large numbers of network parameters under insufficient labeled training samples since the manual labeling process is time-consuming, laborious, and costly. However, some effective methods, such as transfer learning, active learning, and few-shot learning (FSL), are further explored to relieve this problem.

Transfer learning [38] tries to transfer knowledge learned from the source domain to the target domain. Especially, the abundant labeled samples in the source domain ensure that the model parameters are trained appropriately. In addition, the few labeled samples in the target domain can also facilitate further refinement of the trained model. As a seminal work, Yang et al. [39] introduced transfer learning into HSI classification. To solve the problem of domain shift between source and target domain, Li et al. [40] proposed a deep cross-domain FSL (DCFSL) method for HSI classification. Recently, a heterogeneous FSL method was proposed to realize effective knowledge transfer between two heterogeneous datasets [41].

Meanwhile, active learning [42], an excellent supervised learning approach, can filter out more valuable samples. Generally, active learning first utilizes few labeled samples from the training set to train the classifier. Then, it employs a query function to select samples from an unlabeled set. Finally, it labels the chosen samples and expands the initial training set. In [43], an autoencoder combined with active learning was proposed considering rich spectral information in HSI. To accurately describe posterior probability, Deng et al. [44] jointly took spectral and spatial feature representation learning into consideration. Ni and Ma [45] proposed a sparse code histogram (SCH) method to extract more distinguishable features in the active learning procedure.

FSL, different from many other DL methods, aims to explore the difference between samples [46], [47]. Current FSL methods can be classified into three types: the prototype network, the relation network, and the siamese network.

The prototype network calculates the distance between prototype vectors and query samples, and classification is performed with the theory that the samples around the class prototype should belong to the same category. In [48], a novel prototype representation method based on 1-D CNN was proposed to learn the spectral–spatial metric space. Later, to more effectively investigate the spectral–spatial information, a hybrid residual attention module is proposed to build a deep prototypical network [49]. To obtain global prototypical representation, a global prototypical network was proposed with a global representation learning strategy [50].

The relation network, as a metric-based model, is composed of an embedding module based on the prototypical network and the relation module, with a relation module used to calculate the similarity of samples and a learnable module used to describe the difference between samples. In [51], the relation network was first introduced to the HSI classification task. Later, Rao et al. [52] proposed a spatial–spectral relation network (SS-RN) with a 3-D neighborhood to precisely capture the profound similarity between samples. Then, to improve the computational cost and avoid the local optimal solution, a depthwise separable relation network was proposed in [53].

The siamese network is special for its two weight-sharing subnetworks and sample pairs input form, and the Euclidean distance is adopted to determine whether the sample pairs are matched or belong to different categories in the similarity measurement process. In [54], the max-pooling layer and the 2-D convolution layer were used to extract spectral–spatial features. Later, the spectral-spatial siamese network (S3Net) was further exploited to learn deep spatial–spectral features with maximum mean discrepancy (MMD), pairwise loss, and margin-based loss to help the network learn more discriminate features [55]. Cao et al. [56] proposed a siamese network composed of 2-D CNN and 3-D CNN (3DCSN) to jointly train the net with classification loss and contrastive loss. Jia et al. [57] proposed a semisupervised siamese network (3DAES) to fully exploit the supervised and unsupervised information from numerous unlabeled data and limited labeled samples. Recently, we proposed a novel S3Net for few-shot HSI classification, where we innovatively adopted a two-size input form of sample pairs and designed a weighted contrast loss to better identify hard samples [58].

In recent years, graph convolutional networks (GCNs) have received increasing attention due to their remarkable classification performance and excellent ability to model the spatial structure with the arbitrary irregular kernels. Recent studies focused on semisupervised learning, graph construction, pixel-level feature remedy, and attentional mechanism. Qin et al. [59] proposed a semisupervised spatial–spectral GCN (S2GCN) using adjacent nodes in a graph to perform convolution, and the proposed method can make full use of spatial information. To reduce the high computational complexity, Wan et al. [60] performed graph construction on superpixel nodes. However, the graph constructed under superpixel nodes might cause pixel-level feature ignorance to a certain extent. To address this issue, Liu et al. [61] proposed CNN-enhanced GCN (CEGCN). Hong et al. [62] developed a minibatch GCN where large-scale GCNs can be trained in a minibatch fashion with the reduced computational burden. Ding et al. [63] combined spectral information and spatial coordinates to capture the probabilistic connectivity between homogeneous HSI pixel pairs. Later, to better capture the rich spatial contextual information, Ding et al. [64] proposed a global consistent GCN by exploring the topological consistent connectivity. In [65], both spatial and spectral attention blocks were added into GCN to explore an optimal graph through backpropagation during the training process.

The challenges of existing graph convolutional methods are twofold. On the one hand, an inappropriate graph structure is likely to cause oversmoothing in the process of feature extraction; thus, how to obtain a reliable initial graph structure is still a big challenge. On the other hand, how to alleviate the pixel-level feature loss caused by the graph convolutional process based on superpixel nodes is another challenge. Some existing GCN-based methods put forward some solutions. For example, in [60], on the basis of a single-scale segmentation

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

result, the neighbor strategy of different hops is adopted to obtain multiple information when exploring the adjacency relationship between superpixels. In [63], the probability graph is generated by the constantly updated spectral information in the network to adaptively update the graph structure. In addition, in the case of accepting the segmentation results of a single scale, the problem of pixel-level feature loss caused by the unreliable graph structure can be alleviated externally by combining with CNN [61].

Specifically, the drawbacks of existing GCN-based methods using single-scale graph construction are two aspects. For one side, this kind of graph building cannot provide an effective and reliable simulation of the complicated spatial structure of HSI. For another side, a small initial scale tends to form a refined graph, and this graph contains some degree of redundant information and might lose sight of those regions that are spatially distant but highly related in the spectrum. Meanwhile, a bigger initial scale brings a relatively rough graph, which might cause excess smoothness.

In this article, with the motivation of obtaining a reliable initial graph and alleviating the oversmoothing problem in the process of feature extraction, we proposed our solutions internally and externally. For the internal solution, we propose a differentiated-scale GCN, which conducts the graph construction in two different scales to let the features extracted from two graphs complement each other. Meanwhile, we design an adaptive weighted feature fusion strategy with restricted fusion loss to fuse and restrict the differentiated-scale features extracted by the differentiated-scale GCN. For the external solution, to effectively remedy the pixel-level feature, we carry out an integration with our designed lightweight S3Net. As a result, our proposed differentiated-scale restricted GCN (DSR-GCN) can reach a balance between refined localized and rough initial graph structure, and remedy pixel-level feature loss with more distinguishing features learned by the siamese network branch.

Overall, the main contributions of our work are given as follows.

1) We propose a differentiated-scale graph construction method considering both refined localized and rough initial graph structure to better model the spatial structure of HSI.
2) We design restricted fusion loss and an adaptive weighted feature fusion strategy to restrict the fusion of features extracted from differentiated-scale GCN branches.
3) A lightweight S3Net is designed integrated with the differentiated-scale GCN to provide more discriminative pixel-level features.

The main differences between the proposed DSR-GCN method and other GCN-based methods are given as follows.

1) The graph construction in the existing GCN methods [60], [63] is restricted by a single scale. On the contrary, we innovatively construct a graph in differentiate scales, which better leverages the information from smallscale and bigscale graph constructions.
2) Existing GCN methods [61], [62] combined ordinary CNN to remedy pixel-level feature loss. On the opposite, we design a lightweight siamese network to provide more discriminative pixel-level features.

II. Related Work

A. GCNs

The graph is originally used to describe the complex nonlinear structure of non-Euclidean data. To apply it in our task, we need to convert HSI into an undirected graph. In a graph $G(v, e)$, v and e denote the vertex and edges, respectively. In our case, the nodes are representative pixels in the superpixels, and e is the measurement of adjacency among different superpixels.

1) Adjacency Matrix and Laplacian Matrix: The adjacency matrix A in a graph is used to express the adjacency relationships between the graph nodes. In our case, these nodes could be described as representative pixels to those regions derived from the segmentation result, and the radial basis function (RBF) is used to compute and measure the relationship between those representative node vectors. Together with a degree matrix D, whose diagonal elements describes the amount of the number of elements linked to the corresponding nodes in A, the basic characteristic of a graph can be stored. Then, we can compute the symmetric normalized Laplacian matrix L_{sym}

$$L_{\text{sym}} = I - D^{-\frac{1}{2}} A D^{-\frac{1}{2}} \tag{1}$$

where A refers to the adjacency matrix. This way, we can further describe the characteristic of the graph with L_{sym} for the later task.

2) Fourier Transform and Graph Convolution: The Laplacian matrix and the Fourier transformation are important for graph convolution. We can briefly describe the Fourier transformation as a kind of transformation of signal from one domain w to another domain t for the convenience of certain analysis and computation. Through the Fourier transformation, we can change the original convolution to the product in the frequency domain. Thus, graph domain convolution is equivalent to the multiplication of the Fourier domain. Meanwhile, the radical of Fourier transformation is the feature vector of the Laplacian matrix. Then, we can describe the Fourier transformation in the graph domain. To reduce the amount of computation, the Chebyshev polynomial is used to approximate the graph convolution

$$f_\theta(\Lambda) \approx \sum_{k=0}^{K} \theta'_k T_k(\widetilde{\Lambda}) \tag{2}$$

$$U f'_\theta(\Lambda) U^T G = \sum_{k=0}^{K} \theta'_k T_k(\widetilde{L}) G \tag{3}$$

where U and Λ are the eigenvector matrix and the diagonal matrix of eigenvalues of the normalized Laplacian matrix. By limiting this approximation to a first-order approximation, we can conduct the simplified graph convolution as

$$f_\theta * G = \theta\left(\widetilde{D}^{-\frac{1}{2}} \tilde{A} \widetilde{D}^{-\frac{1}{2}}\right) G(k=1) \tag{4}$$

$$\tilde{A} = A + I, \widetilde{D}_{ii} = \sum_j \tilde{A}_{ij} \tag{5}$$

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

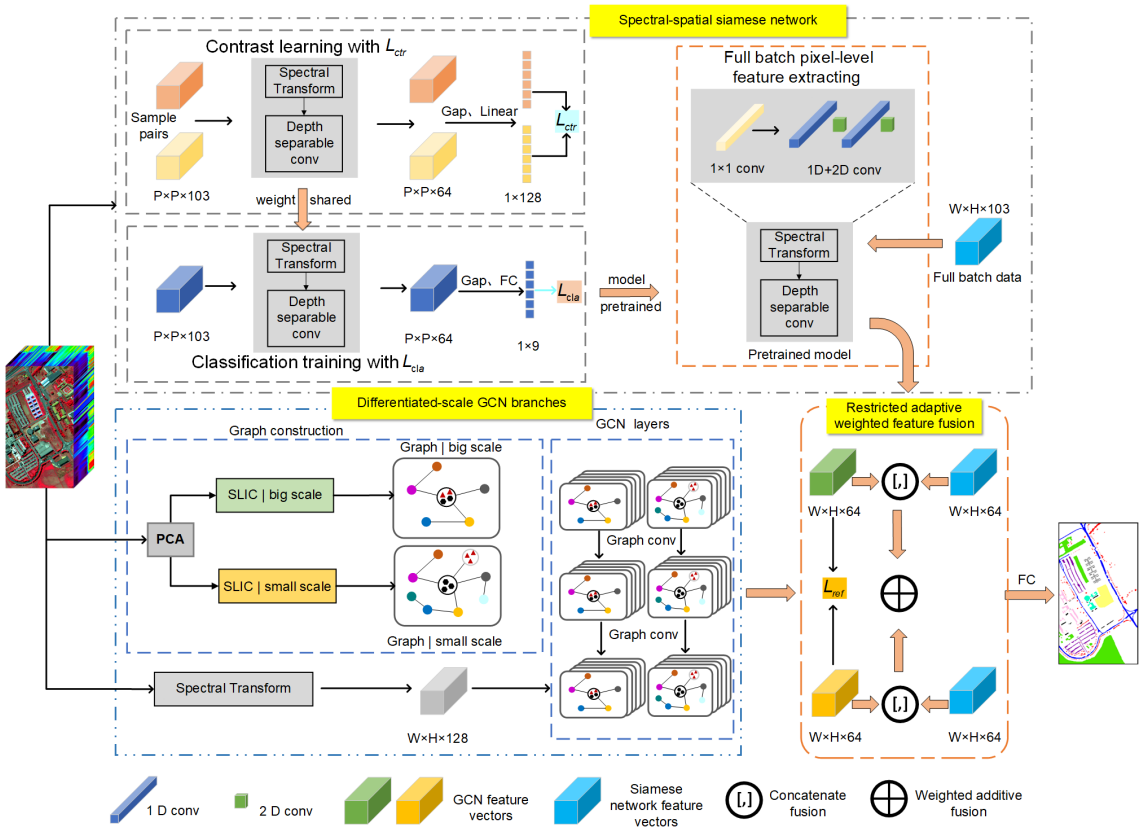


Fig. 1. Architecture of the proposed DSR-GCN.

where G is the vectorization of nodes on the graph. With the activation layer and the weight matrix W at hand, the propagation rule of the graph convolution is

$$H^{l+1} = \sigma\left(\widetilde{D}^{-\frac{1}{2}}\widetilde{A}\widetilde{D}^{-\frac{1}{2}}H^{l}W^{l}\right) \tag{6}$$

where H^l denotes the activated output of the lth layer and W^l is the parameters to be learned. Eventually, with a first-order approximation of spectral filters on the graph at hand, we can carry out the spectral domain graph convolution following this layerwise propagation rule.

B. Siamese Network

A typical siamese network is composed of two parallel subnetworks with the same structure and sharing parameters. The net input is sample pairs, and the subnetworks will map it to a low-dimensional metric space and use the Euclidean distance to measure similarity between sample pairs $\{x_i, x_j\}$

$$d\left(x_i, x_j\right) = \left\|H^{l}(x_i) - H^{l}\left(x_j\right)\right\|_2. \tag{7}$$

However, the traditional siamese network is only able to differ whether the sample pairs belong to the same category, but their specific class is not considered.

To adapt it for the classification task, later siamese networks adopt a manual labeling phase, during which sample pairs are tagged with 1 or 0 determined by whether they belong to the same category or not, and a classification training process is added to the original training stage. Thus, the first stage is contrast learning, which uses a contrast loss term to train the net to learn the difference and similarities between these sample pairs. The contrast loss takes the form

$$L_{\text{ctr}}\left(y_{[0,1]}, d_p, d_n\right) = y_{[1]} \times d_p + \left(1 - y_{[0]}\right) \times \max(\text{margin} - d_n, 0) \tag{8}$$

where $y_{[0,1]}$ denotes the manual label of the sample pairs and $y_{[1]}$ means that the sample pairs are positive pairs, whereas $y_{[0]}$ means that the sample pairs are negative pairs. When the input is positive pairs, the loss term will put a punishment on the distance to each sample. Meanwhile, the loss term will also give encouragement to the distance between negative input pairs. This way, the network will be trained to be capable of narrowing down the intraclass gap and increasing the distance between classes.

In the second training stage, the original samples are fed into the net, and cross-entropy will be used to train the network's classification ability.

III. Proposed Methodology

As shown in Fig. 1, we built our network with three branches. In the first branch, S3Net is used to extract pixel-level features from HSI. The differentiated-scale GCN branches below first apply our graph construction method to encode the original HSI to a rough and refined graph and then export two kinds of corresponding GCN features through the graph convolution on the input. Finally, the features from the three branches will be fused with our feature fusion strategy.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

TABLE I
ARCHITECTURE DETAILS OF THE PROPOSED DSR-GCN (TAKING THE UNIVERSITY OF PAVIA DATASET AS AN EXAMPLE)

Module	Stage	Input size	Layer type	Output size
Spectral-spatial siamese network	Spectral transform	$(103, 9, 9)$	BN, Conv $(Kernelsize = 1, dim = 90)$, ReLU	$(128, 9, 9)$
	Depth separable conv	$(128, 9, 9)$	BN, Conv $(Kernelsize = 1, groups = 1, dim = 128)$, ReLU	$(128, 9, 9)$
		$(128, 9, 9)$	Conv $(Kernelsize = 7, padding = 3, groups = 128, dim = 128)$, ReLU	$(128, 9, 9)$
		$(128, 9, 9)$	BN,Conv $(Kernelsize = 1, groups = 1, dim = 64)$, ReLU	$(128, 9, 9)$
		$(64, 9, 9)$	Conv $(Kernelsize = 7, padding = 3, groups = 64, dim = 64)$,ReLU	$(64, 9, 9)$
	GAP	$(64, 9, 9)$	GlobalAvgPooling	$(64, 4, 4)$
	Linear	$(64, 4, 4)$	Linear $(1024, 128)$	$(1, 128)$
	FC	$(1, 128)$	Linear $(128, 9)$	$(1, 9)$
Differentiated-scale GCN branches	Spectral transform	$(200, 207400)$	BN, Conv $(Kernelsize = 1, dim = 90)$, ReLU	$(90, 207400)$
	GCN layers	$(200, 207400)$	BN, $Linear(dim = 128)$, ReLU	-
		-	BN, Linear $(dim = 64)$, ReLU	$(64, 207400)$
	FC	$(64, 207400)$	Linear $(dim = 9)$	$(9, 207400)$

In addition, the grouping and specific presentation of each module of our network structure are presented in Table I. It can be seen that we group spectral transform, depth separable conv, GAP, linear, and FC to the S3Net module, and we group Spectral transform, GCN layers, and FC to differentiated-scale GCN branches' module.

A. Differentiated-Scale GCN Branches

Generally, to decrease the amount of calculation, the graph building process is usually performed on superpixel nodes rather than pixel nodes. Specific to our classification task under limited samples, we use principal components analysis (PCA) rather than linear discriminant analysis (LDA) to reduce the dimension of HSI before segmentation. Then, simple linear iterative clustering (SLIC) [66] is used to segment HSI into superpixel regions.

In the generating process of the adjacent matrix, we merely use sliding square window to explore the graph topological consistent connectivity in the neighbors according to spatial–spectral similarity between superpixels. Once the square window touches those boundaries, first, we compute the average of all the pixel vectors in each superpixel as follows:

$$X = [X_1, X_2, \ldots, X_c], \quad X_i = \frac{\sum_{k=1}^{n_i} v_k}{n_i} \quad (i = 1 : c) \quad (9)$$

where v_k represents each vector in one superpixel region, n_i is the total number of the vectors it contains, c is the number of unique values of the tag the window covers, and the vector X_i can be regarded as the representative node vector containing averaged information. Then, we use RBF to compute and measure the relationship between these representative node vectors, and the calculated values are stored in adjacency matrix A.

However, when the single segmentation scale is fixed, the sliding window (W) can actually work at the boundaries of superpixels tagged with different numbers, which restricts us from exploring the more refined spatial topological relation and subtle differences in spatial–spectral information within the superpixels. As shown in Fig. 2(a), taking the superpixel blocks numbered 3 and 7 for example, when the scale of these two certain superpixel regions is very big, the surface objects within these superpixels may belong to different classes but are divided into same zone, and this kind of inappropriate segmentation might cause oversmoothing issue.

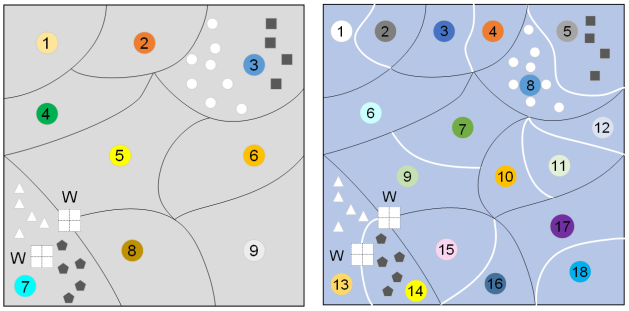


Fig. 2. Graph illustration of the effect of different initial segmentation scales. (a) Large-scale segmentation sketch map. (b) Smallscale resegmentation sketch map.

To address this issue, we use two different initial segmentation scales to perform graph structure construction. First, we choose a suitable initial segmentation scale by conducting experiments on each dataset, and the suitable scale is regarded as a relatively bigscale. Then, we select a relatively smaller scale to build a refined graph on experiment results, which is referred to as smallscale. The smaller scale result can be regarded as a secondary segmentation to relatively rougher segmentation result, which helps to separate those pixels belonging to different classes into separated superpixels, such as superpixel blocks (5, 6) and (13, 14) in Fig. 2(b).

It is worth noting that multiscale graph construction is feasible, but the exploration result shows that it cannot improve the accuracy but increase the burden of the model, which is also one of the reasons why we, finally, choose to use different scales.

B. Restricted Adaptive Weighted Feature Fusion

1) Adaptive Weighted Feature Fusion: The two-scale constructed graph means that we can capture features at different levels together with spatial–spectral pixel-level features extracted by S3Net. To reasonably blend the three kinds of features rather than simply stack them, we add a trainable weight to each part to let the network learn a suitable ratio by itself.

We concatenate the features extracted under a rough graph and relatively refined graph with features captured by the siamese network branch separately, and then, we set

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

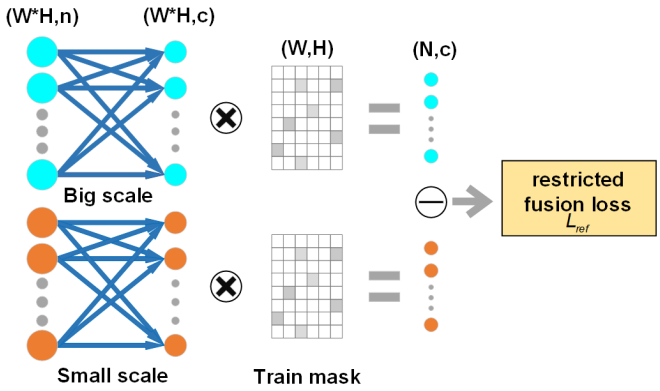


Fig. 3. Graph illustration of restricted fusion loss.

a learnable weight λ and $1-\lambda$ for Concat feature1 and Concat feature2, respectively. We set the initial value of λ to 0.5 so that the features captured by the siamese network are unaffected in the feature fusion stage, and the network adjusts the value of the weight dynamically according to the feedback in the training phase

$$\text{Fusedfeaure} = \lambda \times \text{Concat feature 1} + (1-\lambda) \times \text{Concat feature 2}. \quad (10)$$

2) Restricted Fusion Loss: The strategy used to combine features extracted by GCN under two different scales enables our network to take both relatively rough and refined graph structures into consideration. However, we can also add a restricted fusion loss to the two-part features other than an adaptively weighted combination. Definitely, the features in the hidden layer in the two-scale GCN branches ought to be distinct for the spatial–spectral information; they contain because of the different scale graph. Nonetheless, the two-part features should be consistent when they are projected to the category dimension. Thus, we set a restricted fusion loss term to restrict the feature fusion process.

As shown in Fig. 3, we use the train mask metric, which records the location of training samples, to filter out the corresponding feature vectors projected to the category dimension through the full connected layer. We calculate the restricted fusion loss by square deviation functions as follows:

$$L_{\text{ref}} = \left\| \text{GCN}_{\text{bigscale}}(y, x) - \text{GCN}_{\text{smallscale}}(y, x) \right\|_2^2 \quad (11)$$

where bigscale is the best single scale that we found and smallscale is the other scale smaller than bigscale in the combination of differentiated scales. This loss term can force our network to give close predictions for the same input x with different graphs under bigscale or smallscale.

C. Spectral–Spatial Siamese Network

Commonly, to remedy pixel-level feature loss when we proceed with superpixel-level graph convolution, we combine our base GCN with CNNs. In our proposed method, we further explored this strategy by replacing normal CNNs with S3Net to feed our proposed differentiated-scale GCN with more discriminate pixel-level features.

First, the randomly chosen training samples are expanded to positive and negative sample pairs. Then, the plenty of sample pairs are fed into the contrast leaning phase, where L_{ctr} is used to train the model to distinguish between homogeneous and heterogeneous samples. After that, a cross-entropy loss term L_{cla} is used to perform the classification training

$$L_{\text{cla}}\left(y_{\text{pre}_i}, y_{\text{lab}_i}\right) = \sum_{i=0}^{N} y_{\text{pre}_i} \log\left(y_{\text{lab}_i}\right) \quad (12)$$

where y_{pre_i} represents the predicted output of the ith input samples and y_{lab_i} represents the true label of the ith input samples.

To cooperate with GCN's full batch training mode without adverse impact on the performance of S3Net, we design it in a way that can adapt to both full batch and patch form input data. Then, we separate the training phase into two periods. In the contrast learning and classification training stages, the training data was generated in patch form, and then, full batch data are fed to the network to export full batch pixel-level features, which are next fused with the features extracted by differentiated-scale GCN branches.

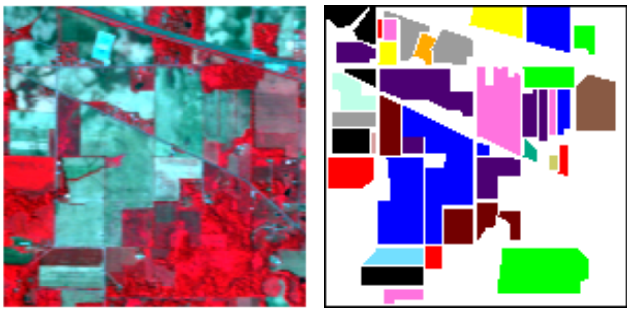
Aiming to avoid the problem of overfitting, we adopt depth separable convolution rather than 3-D convolution, and we use GAP (global average pooling) to replace some of the convolution steps, which helps to reduce net parameters. Thus, our S3Net module can extract more discriminative pixel-level features at low parameter cost, which is difficult to realize with 3-D CNN and normal CNN.

The pseudocode of the DSR-GCN algorithm is shown in Algorithm 1.

Algorithm 1 DSR-GCN

Input: HSI patch train data $\mathbf{X_1}$, HSI full batch data $\mathbf{X_2}$
Step1: Perform PCA on $\mathbf{X_2}$, and use SLIC to generate two scale graphs
Step2: Generate sample pairs $\{x_1, x_2\}$ from $\mathbf{X_1}$
Step3: Apply Spectral transform on sample pairs, $\mathbf{X_1}$
Step4: Perform Contrast learning phase and calculate $\mathbf{L}_{\text{ctr}}$ use Eq. (8) with feature vectors extracted from $\{x_1, x_2\}$
Step5: Perform Classification training phase on $\mathbf{X_2}$ and calculate L_{cla} to the trained siamese network to
Step6: Train siamese network with L_{cla} and L_{cla} and fed $\mathbf{X_2}$ to the trained siamese network to get full batch pixel-level siamese network features
Step7: Get full batch superpixel-level features from differentiated-scale GCN branches
Step8: Perform adaptive feature fusion on the features extracted from three branches using Eq. (10) and calculate L_{ref} using Eq. (11)
Step8: Optimize the loss function and update weights with Adam optimizer
Output: Predict test samples with the trained model

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.



(a) (b)

Fig. 4. Indian Pines. (a) False-color composite image. (b) Ground-truth map.

TABLE II
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR INDIAN PINES

Class	Color	Name	Training	Test
1		Alfalfa	5	41
2		Corn-notill	5	1423
3		Corn-mintill	5	825
4		Corn	5	232
5		Grass-pasture	5	478
6		Grass-trees	5	725
7		Grass-pasture-moved	5	23
8		Hay-windrowed	5	475
9		Oats	5	15
10		Soybeans-notill	5	967
11		Soybeans-mintill	5	2450
12		Soybeans-clean	5	588
13		Wheat	5	200
14		Woods	5	1260
15		Bldg-grass-tree-drivers	5	381
16		Stone-streel-towers	5	88
		Total	80	10169

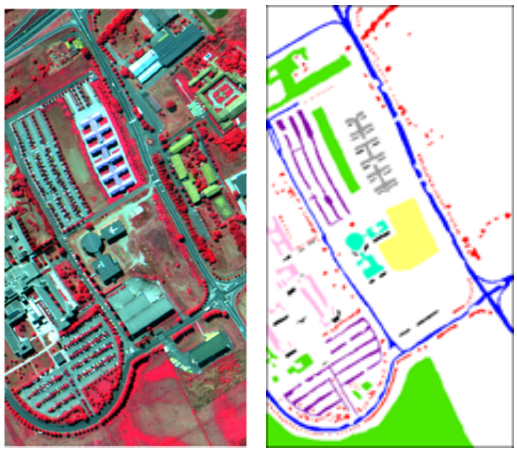
IV. EXPERIMENTS AND RESULTS

In this section, four widely used public HSI datasets were adopted to evaluate the proposed method DSR-GCN: they are the Indian Pines, the University of Pavia, the Salinas and the Houston datasets.

A. Hyperspectral Datasets

More details of the four HSI datasets are given below:

1) *Indian Pines:* The first dataset was acquired by the Airborne Visible/Infrared Imaging Spectrometer AVIRIS) sensor over the Indian Pines test site in Northwestern Indiana. After the removal of the several noisy and water-absorbed bands, the number of wavelength bands in the acquired image is 200 ranging from 0.4 to 2.5 μm, and the image size in the pixel is 145 $\times$ 145, with a moderate spatial resolution of 20 m. The false-color composite of the Indian Pines image and the corresponding ground-truth map are shown in Fig. 4. The ground-truth map totally includes 16 classes. The training and test samples are reported in Table II, where a total of 10 249 samples are available.
2) *University of Pavia:* The second dataset was acquired by the Reflective Optics Spectrographic Imaging System (ROSIS) sensor over the urban area of the University of Pavia, Italy. The number of wavelength bands in the acquired image is 103 ranging from 0.43 to 0.86 μm, and the image size is 610 $\times$ 340, with a very high spatial resolution of 1.3 m. The false-color composite of the University of Pavia image, and the corresponding ground-truth map are shown in Fig. 5. The ground-truth map totally includes nine classes. The training and test samples are reported in Table III, where a total of 42 776 samples are available.



(a) (b)

Fig. 5. University of Pavia. (a) False-color composite image. (b) Ground-truth map.

TABLE III
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR UNIVERSITY OF PAVIA

Class	Color	Name	Training	Test
1		Asphalt	5	6626
2		Meadows	5	18644
3		Gravel	5	2094
4		Trees	5	3059
5		Painted metal sheets	5	1340
6		Bare soil	5	5024
7		Bitumen	5	1325
8		Self-Blocking Bricks	5	3677
9		Shadows	5	942
		Total	45	42731

3) *Salinas:* The third dataset is collected by the AVIRIS sensor in the Salinas area of California, USA. The Salinas dataset has a size of 512 $\times$ 217 and a spatial resolution of 3.7 m per pixel. It contains 204 bands in the spectral dimension, ranging from 400 to 2500 nm. The false-color composite of the Salinas image and the corresponding ground-truth map are shown in Fig. 6. The ground-truth map totally includes 16 classes. The training and test samples are reported in Table IV, where a total of 54 129 samples are available.
4) *University of Houston:* The fourth dataset is collected at the University of Houston campus. The number of wavelength bands in the acquired image is 144 ranging from 0.38 to 1.05 μm, and the image has a size of

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

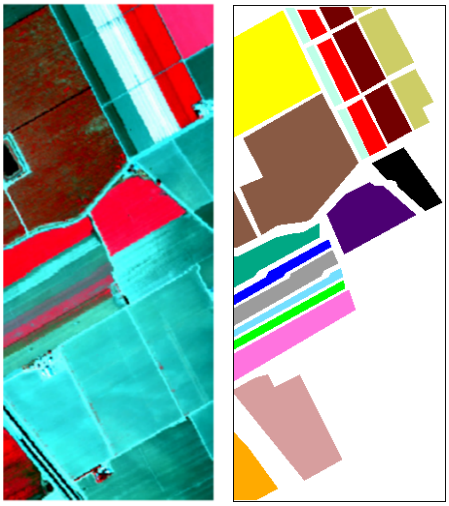


Fig. 6. Salinas. (a) False-color composite image. (b) Ground-truth map.

TABLE IV
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR SALINAS

Class	Color	Name	Training	Test
1		Broccoli green weeds 1	5	2004
2		Broccoli green weeds 22	5	3721
3		Fallow	5	1971
4		Fallow rough plow	5	1389
5		Fallow smooth	5	2673
6		Stubble	5	3954
7		Celery	5	3574
8		Grapes untrained	5	11266
9		Soil vineyard develop	5	6198
10		Corn senesced green weeds	5	3273
11		Lettuce romaine 4wk	5	1063
12		Lettuce romaine 5wk	5	1922
13		Lettuce romaine 6wk	5	911
14		Lettuce romaine 7wk	5	1065
15		Vineyard untrained	5	7263
16		Vineyard vertical trellis	5	1802
		Total	80	54049

349 × 1905, with a high spatial resolution of 2.5 m. The false-color composite of the University of Houston image and the corresponding ground-truth map are shown in Fig. 7. The ground-truth map totally includes 15 classes. The training and test samples are reported in Table V, where a total of 15 029 samples are available.

B. Experimental Settings

1) *For Hyperparameters:* The batch size is set to 16, the training step (epoch) is set to 100, and the learning rate is set to 0.00005, whereas the patch size and the output dimension of spectral transform net and the combination of different numerical values of SLIC's initial segmentation size are analyzed in the flowing experiments.
2) *For Comparison With DSR-GCN:* Several methods have been selected for comparison, including RF [4], 2-D CNN [19], SSRN [21], SSUN [22], DCFSL [40], 3DAES [57], SSFTT [33], ASPCNet [15], and CEGCN [61]. RF is a traditional and baseline method; 2-D CNN

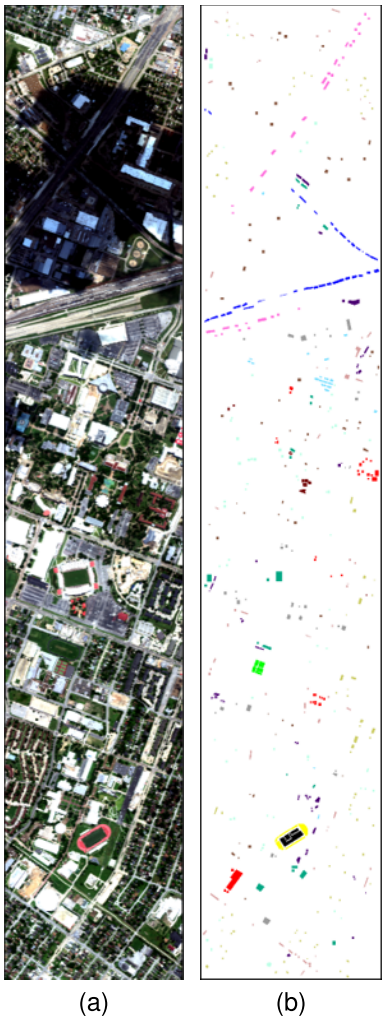


Fig. 7. University of Houston. (a) False-color composite image. (b) Ground-truth map.

TABLE V
LAND CLASSES WITH NUMBER OF TRAINING AND TEST SAMPLES PER CLASS FOR HOUSTON

Class	Color	Name	Training	Test
1		Healthy grass	5	1246
2		Stressed grass	5	1249
3		Synthetic grass	5	692
4		Trees	5	1239
5		Soil	5	1237
6		Water	5	320
7		Residential	5	1263
8		Commercial	5	1239
9		Road	5	1247
10		Highway	5	1222
11		Railway	5	1230
12		Parking Lot1	5	1228
13		Parking Lot2	5	464
14		Tennis court	5	423
15		Running track	5	655
		Total	85	14944

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

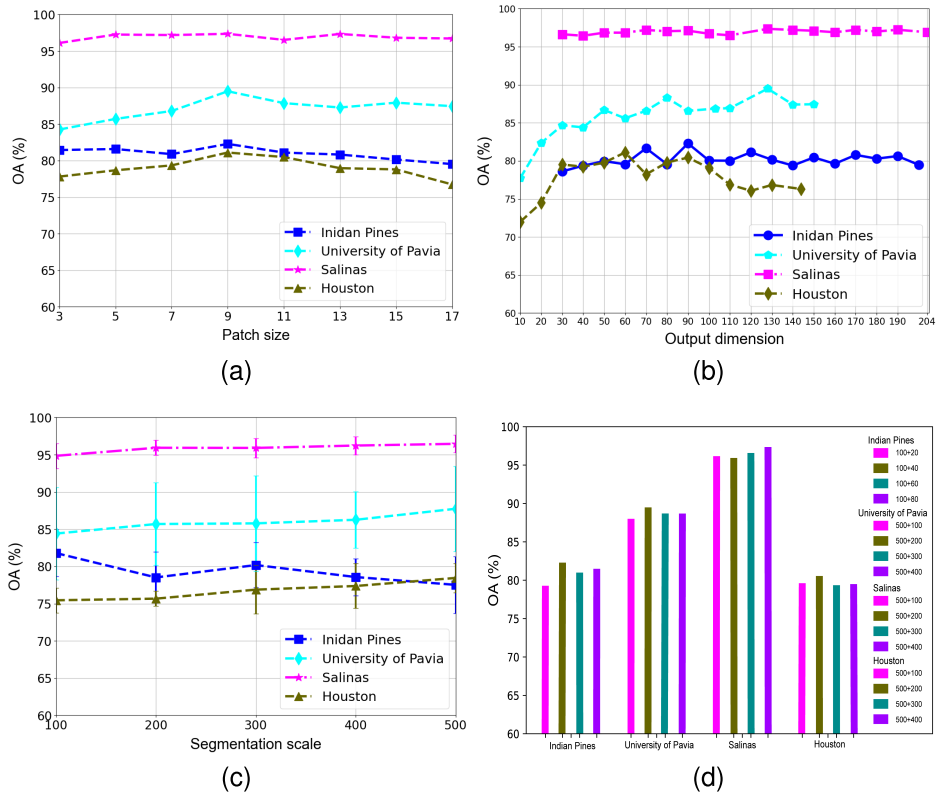


Fig. 8. Evolution of OA as a function of patch size, output dimension, initial segmentation size, and combination of different initial segmentation scales. (a) Patch size. (b) Output dimension. (c) Segmentation scale. (d) Combination of different initial segmentation scales.

is a classic DL method; SSUN and SSRN are start-of-the-art DL methods; DCFSL is a transfer learning method; 3DAES is a start-of-the-art method in siamese neural network sphere; SSFTT is a start-of-the-art transformer method; ASPCNet is a start-of-the-art method in the Capsual network sphere; and CEGCN is a start-of-the-art graph convolutional method. All the methods have adopted the hyperparameters recommended in their original papers. We test these methods under five, ten, 15, 20, 25, and 30 labeled samples per class to evaluate the generalization performance.

3) *Evaluation Metrics:* We select five evaluating metrics to evaluate the experimental results, including overall accuracy (OA), average accuracy (AA), kappa statistics (Kappa), running time (time), and the number of network parameters (parameters).
4) *Running Platforms:* The environment of all experiments comprises Intel[1] Xeon[1] CPU E5-2670 v3 and NVIDIA GeForce RTX 3060. Different DL frameworks are built on Pytorch 1.7.1 in Python 3.6 environment. Note that all experiments results are reported by averaging the results after conducting ten independent runs in terms of randomly selected initial training samples. The source code will be available online: https://github.com/ZhaohuiXue/DSR-GCN.

C. Hyperparameter Analysis for DSR-GCN

To analyze the impacts of patch size, output dimension, initial segmentation size, and the margin value on OA, we set the patch size range from $\{3 \times 3, 5 \times 5, 7 \times 7, 9 \times 9, 11 \times 11, 13 \times 13, 15 \times 15, 17 \times 17\}$ for the four datasets, and the evolution of OA as a function of learning rate is shown in Fig. 8(a). As we can see, when patch size is set to 9×9, OA is generally higher than others. Therefore, we set it to 9×9 in the subsequent experiments.

As for the output dimension of the spectral transform module, we set the range from {30, 40, 50, 60, 70, 80, 90, 100, 110, 128, 140, 150, 160, 170, 180, 190, 200} for the Indian Pines dataset, {10, 20, 30, 40, 50, 60, 70, 80, 90, 103, 110, 128, 140, 150} for the University of Pavia dataset, {30, 40, 50, 60, 70, 80, 90, 100, 110, 128, 140, 150, 160, 170, 180, 190, 204} for the Salinas dataset, and {10,20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120, 130, 144} for the Houston dataset. The evolution of OA as a function of output dimension is shown in Fig. 8(b). As we can see, when the output dimension of spectral transform net is set to 90, 128, 128, and 60, respectively, for the four datasets, we can achieve the best OAs.

Regarding the initial segmentation size, we set it to range from {100, 200, 300, 400, 500} in four datasets, and the evolution of OA as a function of segmentation scale is shown in Fig. 8(c). As we can see, when the segmentation scale is set to 100, 500, 500, and 500, respectively, for the four datasets, the best OAs can be achieved. Then, we optimize an additional smaller initial segmentation size with the relatively bigger optimized initial segmentation size. For Indian Pines, we set the combination range from {(100, 20), (100,40), (100,60), (100, 80)}. For the University of Pavia and Salinas, we set the combination range from {(500, 100), (500,200), (500,300), (500, 400)}. The evolution of OA as a function of

[1]Registered trademark.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION ANALYSIS IN FOUR DATASETS (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Dataset	Baseline	Baseline + SN	Baseline + DS	Baseline + SN + DS	**DSR-GCN (Ours)**
Indian Pines	78.50±3.82	80.40±3.11	80.50±4.00	81.79±3.14	**82.29±3.18**
University of Pavia	85.33±6.56	87.74±5.71	87.66±4.50	88.41±5.55	**89.49±4.21**
Salinas	95.60±1.09	96.46±1.18	96.68±2.26	97.00±1.25	**97.35±0.85**
Houston	76.86±4.00	78.45±1.87	78.94±2.20	80.55±1.73	**81.11±2.52**

Note: Baseline represents GCN with spectral-spatial CNN combined, SN represents siamese network, DS represents differentiated-scale, DSR-GCN represents our method.

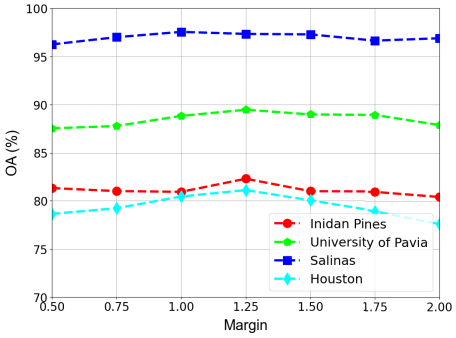


Fig. 9. Evolution of OA as a function of margin.

the combination is shown in Fig. 8(d). As we can see, when the combination of scale is set to (100,40), Indian Pines reaches its highest OA. The optimal combinations for the University of Pavia, Salinas, and Houston are (500, 200), (500, 400), and (500, 200), respectively.

As for the margin value, we set it to range from {0.5, 0.75, 1, 1.25, 1.5, 1.75, 2} in four datasets, and the margin value controls the distance between negative and positive samples. The evolution of OA as a function of learning rate is shown in Fig. 9. As we can see, when the margin value is set to 1–1.25, the model gets the higher OA in the four datasets. In order to make the four datasets can better distinguish the two types of samples, we set the margin value to 1.25.

It is noticed that patch size and segmentation scale are not sensitive to OA on the Salinas dataset. This is because our method has a high and stable classification accuracy on Salinas and benefits from its regular and uncomplicated spatial distribution. Besides, the impact of changes in patch size and segmentation scale on the OA may be weakened during the feature fusion process of the two subnetworks.

D. Ablation Analysis

1) Effectiveness of Siamese Network: To verify the effectiveness of the siamese network, we conduct experiments by comparing GCN with only spectral–spatial CNN branch (Baseline) attached and GCN with the S3Net branch attached (Baseline + SN) with five labeled samples per class, and the averaged OAs are reported in Table VI. The OA result is raised by 1.9%, 2.41%, 0.86%, and 1.59% on four datasets under five samples, respectively, which demonstrates the benefits of integrating the siamese network with our base network on the OA performance. In other words, the additional contrast learning phase helps our base network to seize more useful features and information from given limited samples.

2) Effectiveness of Differentiated Scale: To verify the effectiveness of differentiated scale, we conduct experiments by comparing GCN with only spectral–spatial CNN branch (Baseline) attached and GCN with differentiated-scale (Baseline + DS) under five labeled samples per class, and the averaged OAs are reported in Table VI. The OA result is raised by 2.0%, 2.33%, 1.08%, and 2.08% on four datasets under five samples, respectively, which demonstrates the benefits of graph construction with the differentiated scale on the OA performance.

3) Adaptive Weighted Differentiated-Scale Feature Fusion Strategy: To verify the effectiveness of the adaptive weighted two-scale feature fusion strategy, we conduct experiments by comparing using only one optimal segmentation scale (Baseline + SN) and using the best combination of two differentiated scales (Baseline + SN + DS) for each dataset under five labeled samples per class, and the averaged OAs are reported in Table VI. The OA results are raised by 1.39%, 0.67%, 0.54%, and 2.10% for the four datasets, respectively. The result shows the effectiveness of our strategy to use adaptive weighted feature fusion to rationally integrate features extracted under differentiated scales. Normally, the roughness of the GCN superpixel-level features can be solved or relieved by feature fusion with CNN pixel-level features, we thought that the GCN itself can alleviate its own problems through the fusion of features extracted under differentiated scales, and the result again proves it to be reasonable.

4) Restricted Fusion Loss Term: To validate the effectiveness of the restricted fusion loss, we attach it to the differentiated-scale feature fusion process (DSR-GCN). As shown in Table VI, the OA results are increased by 0.5%, 1.08%, 0.35%, and 0.56% on four datasets, respectively. The result proves that L_{ref} helps our network to rationally fuse differentiated-scale GCN features, which ensures that the differentiated-scale features are mutually remedying rather than contradictory.

E. Generalization Performance

To validate the generalization performance of DSR-GCN, we compare the classification results of different methods under limited training samples. For the four datasets, we randomly select the labeled samples per class according to the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

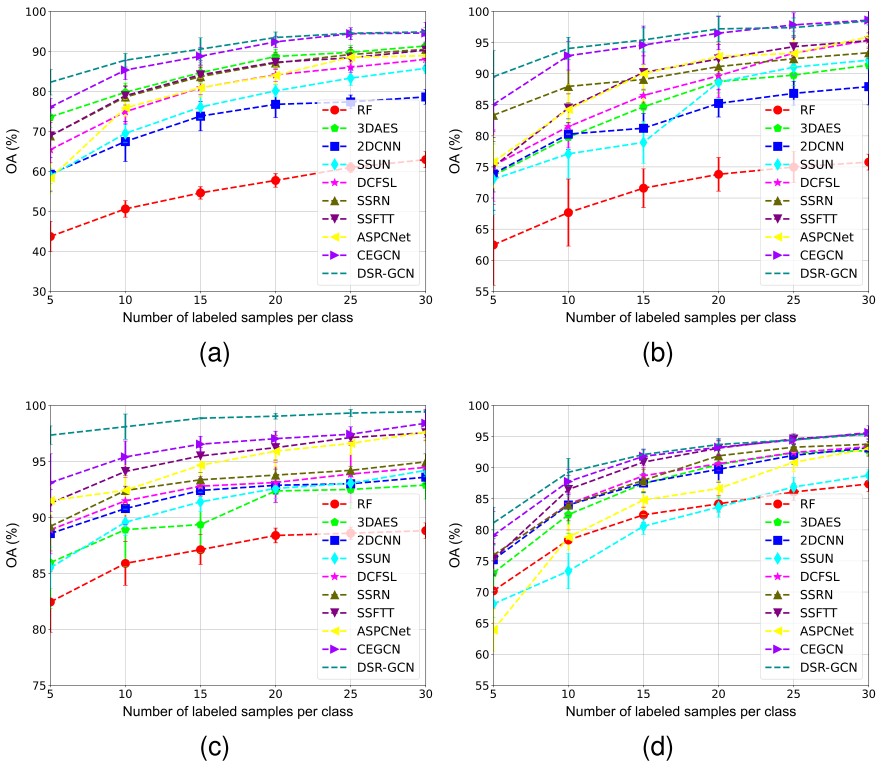


Fig. 10. Evolution of OA as a function of the number of labeled samples per class. (a) Indian Pines. (b) University of Pavia. (c) Salinas. (d) Houston.

TABLE VII
CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR INDIAN PINES (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	RF [4]	2D-CNN [19]	SSRN [21]	SSUN [22]	DCFSL [40]	3DAES [57]	SSFTT [33]	ASPCNet [15]	CEGCN [61]	**DSR-GCN(Ours)**
1	59.02±14.09	91.95±12.2	54.62±27.81	**100.0±0.0**	94.88±7.51	99.35±1.94	99.76±0.73	31.16±11.92	97.56±2.43	98.04±1.82
2	21.20±6.78	33.60±13.30	**77.57±15.07**	40.98±11.90	42.30±10.19	47.05±10.91	53.09±11.08	53.44±6.78	54.38±11.56	69.78±8.75
3	37.61±9.46	35.73±11.42	64.32±14.09	63.71±11.96	49.24±8.89	47.24±6.88	56.76±11.23	52.85±11.26	61.59±9.29	**78.55±14.68**
4	29.78±6.75	59.72±14.55	52.50±21.96	78.23±16.49	74.40±8.34	74.91±8.48	**91.21±7.01**	30.04±6.10	85.64±11.75	89.43±10.49
5	56.72±20.75	79.13±10.65	**86.03±15.66**	63.35±11.76	73.81±6.44	82.56±5.97	81.62±8.41	61.42±13.07	78.24±7.66	77.17±7.58
6	75.21±10.85	93.67±3.46	87.58±16.82	60.62±10.98	88.08±4.79	83.26±6.75	91.84±6.6	73.33±10.35	**96.66±2.54**	94.75±3.07
7	83.91±10.48	97.83±7.08	54.15±20.11	**100.0±0.0**	99.13±1.74	**100.0±0.0**	**100.0±0.0**	36.64±24.89	**100.0±0.0**	**100.0±0.0**
8	63.02±16.66	93.96±9.09	98.17±2.08	96.91±3.11	85.90±9.48	95.75±1.97	94.35±4.7	85.50±8.84	96.27±7.02	**99.21±2.27**
9	82.0±14.0	99.00±4.36	38.28±21.20	**100.0±0.0**	98.67±4.0	**100.0±0.0**	98.67±4.0	9.08±4.08	**100.0±0.0**	98.00±4.26
10	32.34±8.94	56.37±12.91	66.32±17.31	44.33±10.44	59.79±9.65	59.52±11.78	67.18±8.08	48.54±7.45	68.25±13.68	**73.38±10.59**
11	34.83±11.17	52.36±12.29	76.00±10.87	51.71±12.55	61.49±11.68	55.03±8.99	57.04±12.67	74.52±6.69	75.41±8.92	**80.51±12.38**
12	25.60±5.2	31.85±7.62	52.09±11.18	49.69±10.97	47.53±14.18	51.09±8.72	55.94±10.12	59.37±12.85	63.04±9.97	**70.27±9.79**
13	90.45±5.90	96.75±3.18	81.64±22.39	97.65±2.18	97.55±3.27	**99.68±0.67**	96.65±5.0	51.49±13.44	99.45±0.98	98.30±1.41
14	73.57±14.81	81.87±11.38	90.15±17.19	69.50±17.25	86.28±8.7	87.89±3.72	82.86±7.92	81.64±9.30	91.62±7.02	**92.02±7.48**
15	22.41±7.02	56.06±17.68	76.57±16.45	71.76±17.39	69.66±8.1	74.88±7.95	75.52±10.66	54.83±14.73	86.69±9.77	**99.02±1.45**
16	94.89±3.02	99.49±1.11	70.76±16.67	97.05±4.61	98.41±1.92	99.62±1.15	99.09±2.08	38.04±14.71	98.52±2.73	**99.77±0.45**
OA (%)	43.71±3.77	59.30±4.24	68.88±3.82	58.88±1.91	65.43±3.23	65.37±1.63	68.88±3.39	58.54±3.80	76.09±2.18	**82.29±3.18**
AA (%)	55.16±1.23	72.46±2.17	70.42±4.56	74.09±1.42	76.69±2.33	78.61±1.00	81.35±1.91	52.62±4.42	84.58±1.59	**88.64±1.8**
$\kappa\times100$	37.43±3.61	54.56±4.37	65.11±3.99	54.08±1.95	61.0±3.49	60.99±1.76	65.18±3.58	54.05±4.02	72.88±2.42	**79.94±3.58**

range of [5, 10, 15, 20, 25, 30], and the OAs as a function of the number of labeled samples per class on four datasets are given in Fig. 10. As we can see, DSR-GCN gets the highest accuracies in most cases for different datasets, which proves the superiority in terms of the generalization performance of our method. It is worth mentioning that DSR-GCN gets prominent superiority over the Salinas dataset; we consider that this dataset is relatively large and samples are more dispersed; and both the siamese network's advantages in small samples and the differentiated initial segmentation scale strategy are beneficial to the performance improvements.

F. Classification Results

To validate the superiority of the proposed DSR-GCN, we compare RF, 2-D CNN, SSUN, SSRN, DCFSL, 3DAES, SSFTT, ASPCNet, CEGCN, and our method DSR-GCN using five labeled samples per class in four datasets, and the averaged results are reported in Tables VII–X. To visually inspect the classification performance, we exhibit the classification maps obtained by different methods in Figs. 11–14, respectively.

According to the result of Indian Pines reported in Table VII, DSR-GCN surpasses the other methods obviously with the highest accuracies. Specifically, DSR-GCN obtains the highest OA of 82.29 ± 3.18%, which is 6.20%–38.58% higher than the other methods. In addition, DSR-GCN also achieves significant augments in terms of AA and Kappa compared to others. Moreover, DSR-GCN gets the highest accuracies in nine of 16 classes, which also proves our method's superiority.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

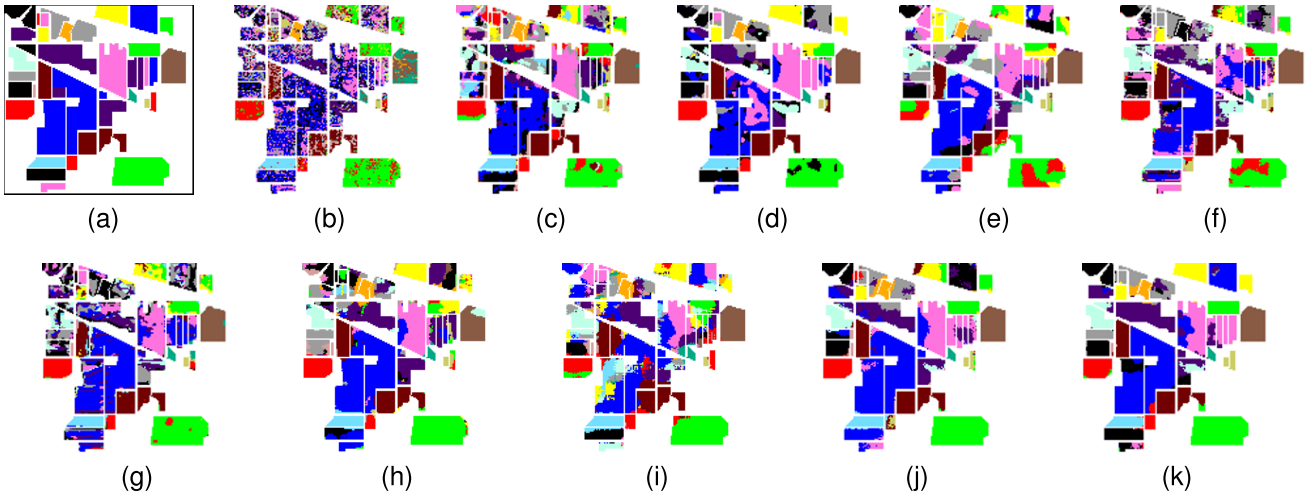


Fig. 11. Classification maps of different methods for Indian Pines (five labeled samples per class for training). (a) Ground truth. (b) RF (43.71%). (c) 2-D CNN (59.30%). (d) SSRN (68.88%). (e) SSUN (58.88%). (f) DCFSL (65.43%). (g) 3DAES (65.37%). (h) SSFTT (68.88%). (i) ASPCNet (58.54%). (j) CEGCN (76.09%). (k) DSR-GCN (82.29%).

TABLE VIII

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR UNIVERSITY OF PAVIA (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	RF [4]	2D-CNN [19]	SSRN [21]	SSUN [22]	DCFSL [40]	3DAES [57]	SSFTT [33]	ASPCNet [15]	CEGCN [61]	**DSR-GCN (Ours)**
1	49.18±10.64	50.03±18.37	79.42±11.96	67.42±6.15	77.89±15.63	69.02±13.08	67.52±12.63	81.72±6.64	83.74±5.93	**86.39±7.69**
2	62.14±16.78	77.06±12.94	89.35±5.21	73.93±15.93	86.40±9.13	81.55±7.37	68.97±9.34	**94.75±3.24**	80.45±10.15	90.73±8.85
3	45.90±14.81	87.14±9.26	72.44±9.88	46.35±7.70	63.97±9.73	54.97±11.66	79.84±7.93	51.92±10.60	90.00±10.54	**96.39±3.39**
4	88.34±6.89	84.69±7.02	79.78±10.62	87.78±6.58	**91.74±5.21**	90.18±5.67	81.22±7.79	67.22±13.53	90.34±5.59	85.96±4.48
5	99.80±0.11	99.94±0.12	99.67±0.29	86.96±8.02	99.25±0.78	98.10±2.47	98.5±1.97	88.54±2.68	99.88±0.18	**99.99±0.02**
6	51.31±10.64	76.35±14.51	67.39±21.58	72.70±19.07	75.34±8.87	48.90±6.44	89.78±5.07	65.89±15.21	89.49±11.56	**94.19±7.347**
7	73.07±13.55	98.86±1.16	67.23±13.91	69.41±6.80	78.06±8.66	70.83±18.92	98.6±1.17	60.88±6.47	**99.89±0.13**	99.17±1.23
8	64.07±14.95	46.17±21.50	57.41±15.03	83.91±6.80	64.56±12.14	59.82±7.44	69.27±16.06	58.57±9.67	84.64±15.84	**85.79±11.67**
9	**99.84±0.340**	97.54±2.86	84.29±10.95	89.61±8.96	98.61±2.57	98.86±0.77	94.65±4.6	50.42±13.29	92.47±6.81	86.60±12.95
OA (%)	62.46±6.48	73.01±5.68	75.23±5.75	73.89±5.85	81.60±3.90	73.61±2.61	75.04±4.07	75.83±4.21	85.04±4.30	**89.49±4.21**
AA (%)	70.41±2.33	79.75±3.36	75.34±3.33	77.44±3.37	81.76±2.61	74.69±3.66	83.15±2.50	68.88±3.85	90.10±1.95	**91.01±2.35**
$\kappa\times100$	53.27±6.37	65.92±6.53	66.77±6.15	68.23±6.36	76.15±4.56	65.87±3.40	68.93±4.36	68.98±4.97	80.94±5.01	**86.39±5.18**

TABLE IX

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR SALINAS (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	RF [4]	2D-CNN [19]	SSRN [21]	SSUN [22]	DCFSL [40]	3DAES [57]	SSFTT [33]	ASPCNet [15]	CEGCN [61]	**DSR-GCN (Ours)**
1	99.09±0.70	98.91±1.50	99.28±1.70	77.70±14.91	99.61±0.35	99.39±1.20	99.49±0.89	95.84±5.25	**99.88±0.31**	99.31±1.95
2	93.95±5.60	99.59±1.01	97.52±3.46	74.32±9.18	99.35±1.66	88.93±19.80	99.84±0.2	99.93±0.09	**100.00±0.00**	**100.00±0.00**
3	87.92±6.94	96.31±6.15	99.53±0.80	86.77±6.19	99.35±1.66	98.44±2.36	99.83±0.44	98.68±1.60	**100.00±0.00**	**100.00±0.00**
4	99.18±0.41	96.79±2.46	93.01±4.12	99.93±0.07	99.08±0.95	99.35±0.38	95.84±5.63	91.88±7.95	**99.92±0.08**	96.22±0.60
5	**99.86±2.71**	92.07±4.73	96.51±3.33	90.79±9.51	92.75±3.16	95.59±2.56	96.38±2.93	97.46±2.00	96.92±0.33	95.04±0.81
6	99.39±0.27	98.19±2.02	98.61±1.31	98.24±3.25	99.58±0.78	99.79±0.22	98.28±1.8	98.64±1.48	**99.94±0.10**	99.83±0.15
7	99.68±0.19	99.02±1.67	97.66±3.66	90.49±10.63	98.83±2.53	99.61±0.45	99.46±0.74	99.59±0.62	**99.93±0.06**	99.85±0.07
8	62.18±15.98	75.89±11.79	79.27±8.56	66.83±9.39	73.54±12.60	67.19±15.10	75.6±9.46	86.07±4.53	80.80±12.46	**97.71±0.30**
9	98.86±0.98	99.78±0.37	96.93±4.93	99.43±0.60	99.12±1.15	97.10±2.42	99.97±0.03	97.15±2.63	**100.0±0.0**	**100.00±0.00**
10	80.20±6.20	87.81±4.967	92.94±8.45	87.89±9.98	83.90±12.12	90.90±2.95	89.01±12.65	**93.62±3.75**	91.19±3.57	85.50±10.21
11	93.19±1.65	98.91±1.78	95.17±2.66	95.76±2.70	99.22±0.85	99.16±0.63	**99.91±0.13**	96.23±6.09	99.84±0.33	99.79±0.41
12	91.04±4.24	97.62±2.78	98.79±1.99	96.38±4.00	99.58±0.41	99.15±1.59	99.26±0.59	96.28±3.84	**99.64±0.82**	99.53±0.83
13	90.35±6.730	97.33±5.95	90.39±10.63	99.45±0.44	99.20±0.47	99.66±0.35	97.98±4.07	89.31±8.35	**100.00±0.00**	94.28±11.60
14	78.08±15.95	94.63±10.09	86.02±10.87	98.79±0.76	98.55±0.93	97.75±1.61	96.13±2.51	94.17±5.93	**99.28±0.66**	84.79±9.33
15	55.16±13.61	68.60±11.43	73.67±9.46	86.38±6.59	77.49±7.64	72.22±20.74	82.85±11.49	77.33±11.16	84.62±11.85	**97.02±2.84**
16	95.71±2.05	91.00±6.76	97.39±3.21	87.34±5.56	92.70±5.82	96.63±3.02	98.27±1.02	**100.00±0.00**	96.99±4.64	97.36±6.36
OA (%)	82.44±2.70	88.56±2.76	88.89±2.14	85.50±1.68	89.20±2.20	87.17±1.67	91.26±1.23	91.50±1.23	93.10±2.58	**97.35±0.82**
AA (%)	88.80±1.48	93.28±1.36	93.62±1.47	89.78±1.08	93.98±2.20	93.80±1.44	95.51±0.95	94.51±0.97	**96.81±1.02**	96.64±0.91
$\kappa\times100$	80.51±2.92	87.29±3.03	87.62±2.40	83.94±2.04	88.01±2.41	85.76±1.85	90.30±1.35	90.55±1.34	92.33±2.85	**97.05±0.91**

According to the result of the University of Pavia reported in Table VIII, DSR-GCN outperforms the other methods obviously with the highest accuracies. Specifically, DSR-GCN obtains the highest OA of 89.49 ± 4.21%, which is 4.45%–27.03% higher than the other methods. In addition, DSR-GCN also produces significant improvements in terms of AA and Kappa compared to others. As for the class-specific accuracy, our method achieves the highest accuracies in five of nine classes, which demonstrates the superiority of DSR-GCN.

According to the result of Salinas reported in Table IX, DSR-GCN outperforms the other methods obviously with the highest accuracies. Specifically, DSR-GCN obtains the highest OA of 97.35 ± 0.82%, which is 4.25%–14.91% higher than the other methods. In addition, DSR-GCN also gets a far better classification map with higher homogeneity within class and better separability between classes. It is noticed that the AA of our method is lower than CEGCN, which is mainly because the classification performance of our method in the 13th and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

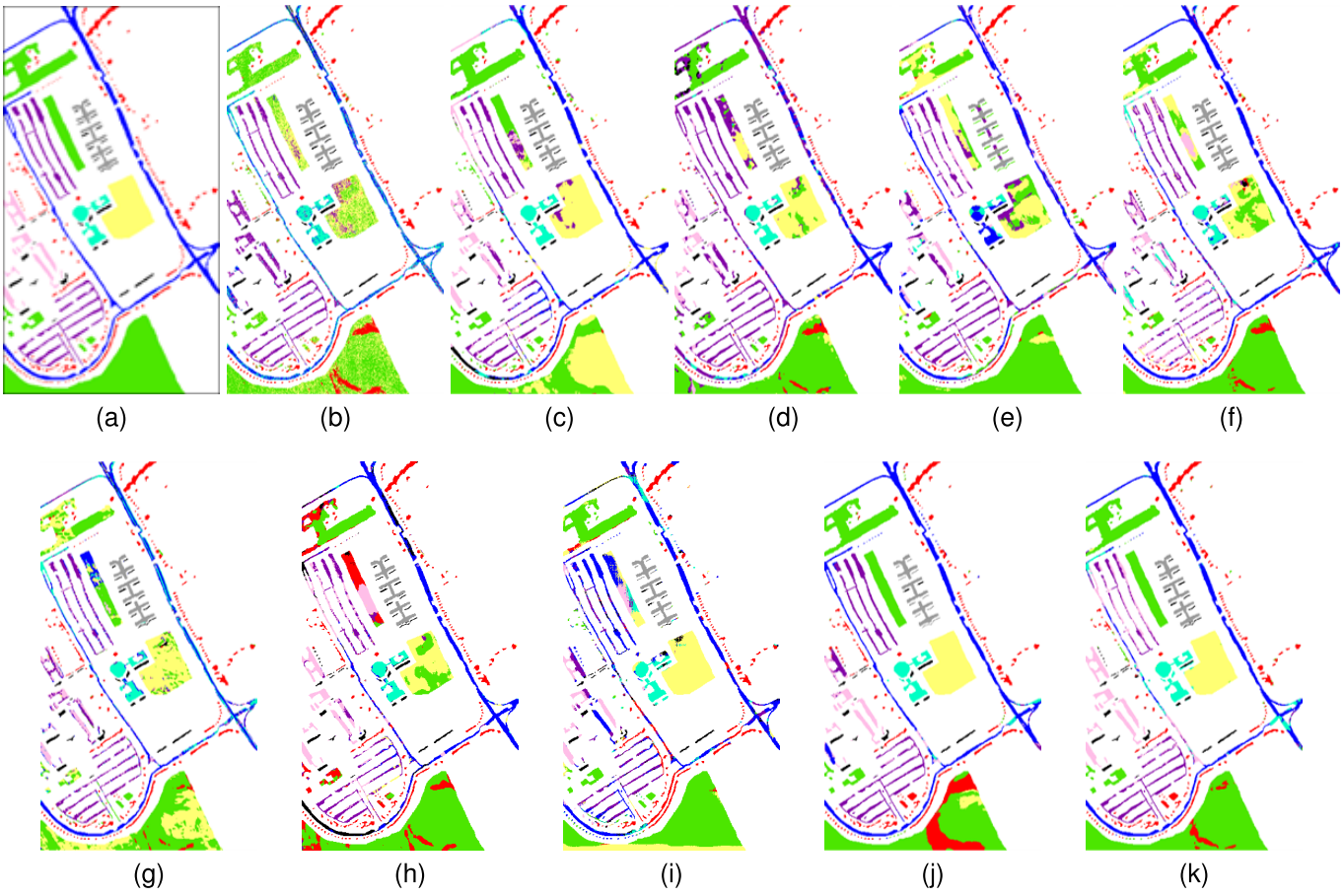


Fig. 12. Classification maps of different methods for the University of Pavia (five labeled samples per class for training). (a) Ground truth. (b) RF (62.46%). (c) 2-D CNN (73.01%). (d) SSRN (73.89%). (e) SSUN (75.23%). (f) DCFSL (81.60%). (g) 3DAES (73.61%). (h) SSFTT (75.04%). (i) ASPCNet (75.83%). (j) CEGCN (85.04%). (k) DSR-GCN (89.49%).

TABLE X

CLASSIFICATION ACCURACIES OF DIFFERENT METHODS FOR HOUSTON (FIVE LABELED SAMPLES PER CLASS FOR TRAINING)

Class	RF [4]	2D-CNN [19]	SSRN [21]	SSUN [22]	DCFSL [40]	3DAES [57]	SSFTT [33]	ASPCNet [15]	CEGCN [61]	**DSR-GCN(Ours)**
1	79.55±7.77	84.71±9.29	83.75±7.79	88.40±9.85	88.92±6.40	**91.27±6.42**	79.79±6.99	66.79±16.60	88.53±5.82	85.20±6.86
2	90.15±6.00	88.61±5.95	67.01±11.75	**91.08±6.09**	79.03±11.23	73.61±15.97	86.21±7.22	64.07±10.88	68.80±9.00	87.45±5.97
3	**100.00±0.00**	91.21±5.06	93.89±4.18	97.06±5.36	96.53±3.19	96.79±3.19	93.88±5.58	83.22±5.31	98.90±1.11	99.19±0.57
4	87.65±2.86	88.76±5.64	77.56±13.90	89.75±7.73	90.79±3.21	88.52±6.31	82.54±7.18	58.66±14.98	89.77±5.54	**92.14±4.36**
5	95.60±5.83	**99.54±0.79**	89.14±7.69	91.57±4.90	95.24±3.80	86.24±9.13	98.14±1.96	85.02±8.64	94.74±7.86	97.42±4.51
6	89.56±4.45	89.78±4.86	63.62±18.86	**95.51±4.75**	80.97±6.58	84.09±4.57	84.86±5.64	71.46±16.15	87.59±4.47	74.91±14.27
7	60.68±14.48	47.87±16.13	75.53±4.59	74.62±15.58	63.06±13.90	63.46±10.45	58.52±11.13	49.60±6.21	**81.14±12.27**	80.89±4.86
8	41.01±8.14	47.99±10.35	33.43±7.57	**86.42±11.32**	44.67±5.01	58.10±7.29	44.07±9.99	69.58±10.84	41.84±9.56	44.06±12.71
9	56.46±9.40	50.44±9.23	76.78±10.82	59.62±12.02	63.18±9.36	64.02±8.42	54.78±8.93	44.03±12.14	76.37±8.48	**81.64±9.01**
10	57.19±8.59	66.51±12.39	48.63±12.80	54.52±10.51	66.04±14.39	54.49±10.05	65.73±12.7	57.62±10.25	78.26±19.14	**78.55±12.29**
11	64.44±9.33	74.91±8.17	58.13±8.37	58.71±12.85	62.93±8.75	56.40±12.12	76.74±11.62	69.02±9.98	**77.37±10.49**	64.84±15.27
12	44.50±10.87	67.52±10.06	55.45±11.41	71.98±10.55	73.66±11.54	62.75±13.37	67.87±9.82	70.00±14.69	70.07±12.03	**75.81±13.62**
13	33.92±3.79	81.10±13.92	39.40±7.76	70.07±12.74	86.66±7.66	68.03±9.67	**88.52±6.67**	42.95±9.10	70.95±14.23	62.61±20.94
14	83.92±15.54	99.22±1.73	91.56±5.63	91.99±6.48	88.30±8.13	88.35±7.43	**99.95±0.09**	88.89±13.48	97.16±1.91	98.25±2.46
15	98.99±0.30	98.72±2.50	76.02±17.66	93.11±5.00	95.73±2.57	95.97±6.13	99.19±1.38	77.40±9.54	**99.63±0.71**	97.53±3.97
OA (%)	70.49±2.77	75.20±2.50	75.93±3.00	68.08±2.14	75.91±2.67	73.04±4.75	75.30±2.54	63.88±3.38	79.11±3.82	**81.11±2.52**
AA (%)	72.24±2.60	78.46±2.39	68.66±2.14	80.96±2.24	78.38±2.41	75.47±4.05	78.72±2.32	66.55±3.01	81.21±3.41	**82.40±1.91**
$\kappa\times100$	68.11±2.99	73.23±2.69	65.54±2.31	74.00±3.24	73.97±2.88	70.89±5.10	73.32±2.74	61.03±3.67	77.41±4.14	**79.57±2.72**

14th categories was much lower than that of CEGCN. However, our method is superior to CEGCN in overall performance.

According to the result of Houston reported in Table X, DSR-GCN outperforms the other methods obviously with the highest accuracies. Specifically, DSR-GCN obtains the highest OA of 81.11 ± 2.52%, which is 2.0%–17.23% higher than the other methods. In addition, DSR-GCN also produces significant improvements in terms of AA and Kappa compared to others, which demonstrates the superiority of DSR-GCN.

V. DISCUSSION

A. Feature Separability

To display the separability of different feature maps, we use PCA to transform the features maps into 2-D space, as shown

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

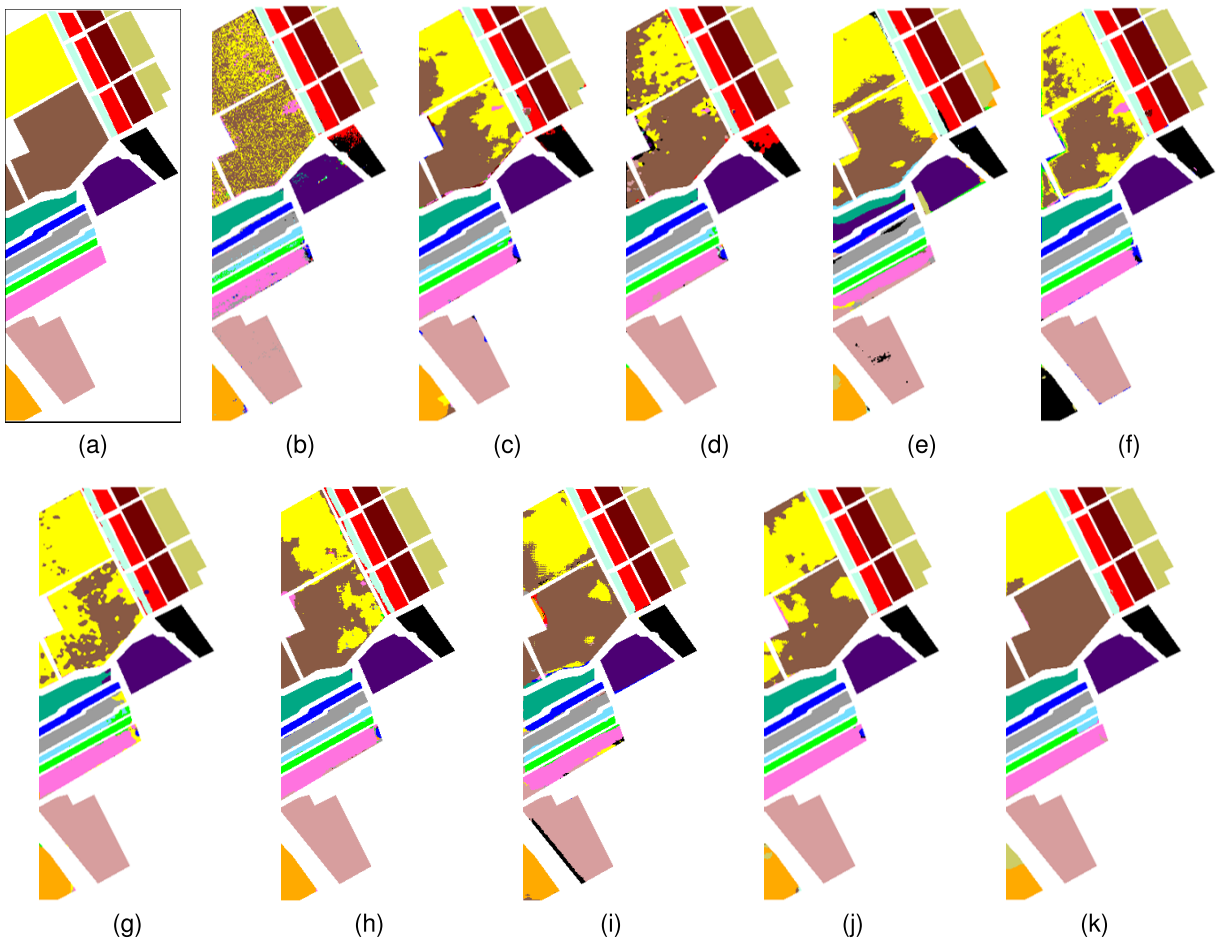


Fig. 13. Classification maps of different methods for Salinas (five labeled samples per class for training). (a) Ground truth. (b) RF (82.44%). (c) 2-D CNN (88.56%). (d) SSRN (88.89%). (e) SSUN (85.50%). (f) DCFSL (89.20%). (g) 3DAES (87.17%). (h) SSFTT (91.26%). (i) ASPCNet (91.50%). (j) CEGCN (93.10%). (k) DSR-GCN (97.35%).

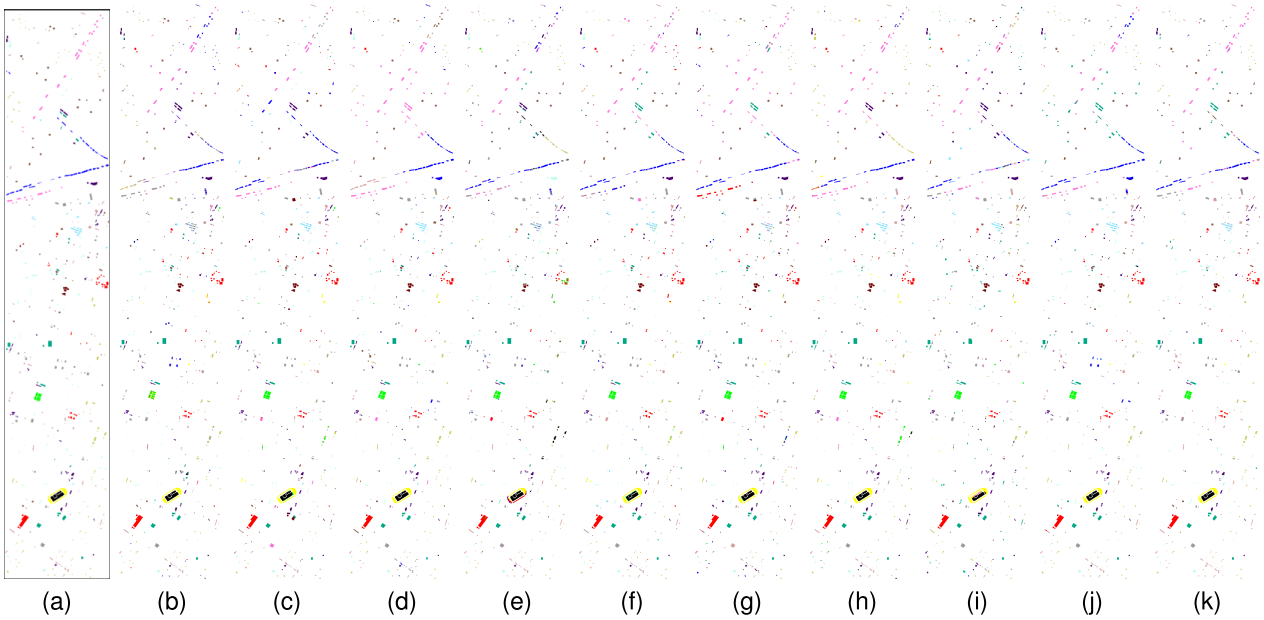


Fig. 14. Classification maps of different methods for Houston (five labeled samples per class for training). (a) Ground truth. (b) RF (70.49%). (c) 2-D CNN (75.20%). (d) SSRN (75.93%). (e) SSUN (68.08%). (f) DCFSL (75.91%). (g) 3DAES (73.04%). (h) SSFTT (75.30%). (i) ASPCNet (63.88%). (j) CEGCN (79.11%). (k) DSR-GCN (81.11%).

in Fig. 15. Compared with the original features, the features extracted by our method are well separated, and it can be clearly seen that the feature separability of the DSR-GCN is better than Baseline, which demonstrates that the S3Net

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

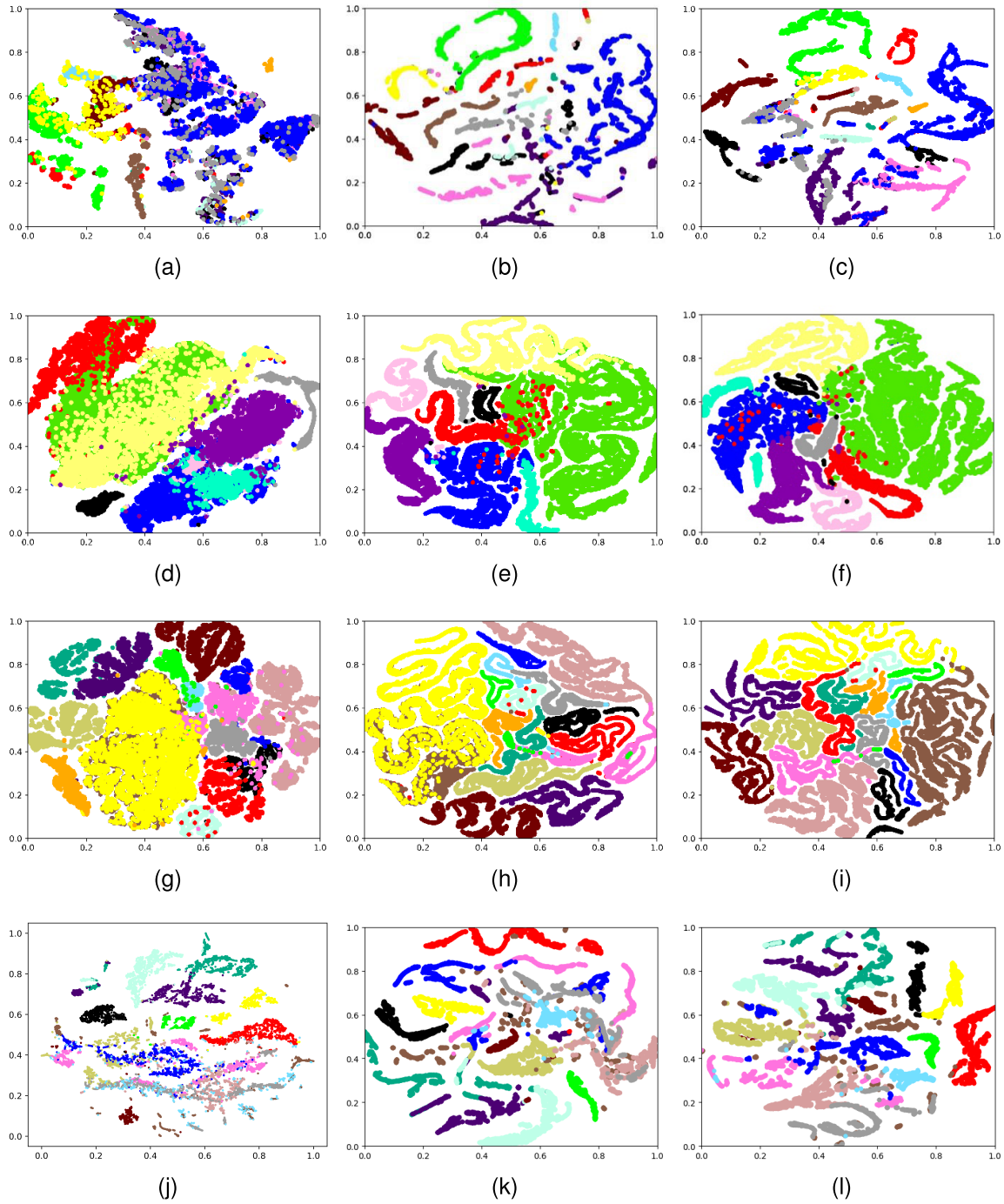


Fig. 15. Feature separability obtained by DSR-GCN for four datasets. (a) Raw feature for Indian Pines. (b) Baseline and (c) DSR-GCN for Indian Pines. (d) Raw feature for the University of Pavia. (e) Baseline and (f) DSR-GCN for the University of Pavia. (g) Raw feature for Salinas. (h) Baseline and (i) DSR-GCN for Salinas. (j) Raw feature for Houston. (k) Baseline and (l) DSR-GCN for Houston. (a) $S_b = 0.17$. (b) $S_b = 1.31$. (c) $S_b = 1.5$. (d) $S_b = 0.06$. (e) $S_b = 0.91$. (f) $S_b = 1.07$. (g) $S_b = 0.04$. (h) $S_b = 1.59$. (i) $S_b = 1.8$. (j) $S_b = 0.01$. (k) $S_b = 0.86$. (l) $S_b = 1.0$.

branch of DSR-GCN can increase the distance between classes through the contrast learning stage.

In addition, we also calculate the interclass dispersion coefficient in the Fisher discriminant analysis to further quantitatively analyze the feature separability of our method. The interclass dispersion coefficient can be calculated as

$$S_b = \sum_{i=1,\, j=1,\, i\neq j}^{c} \left(m_i - m_j\right)\left(m_i - m_j\right)^T/(c-1) \qquad (13)$$

where m_i and m_j are the mean vectors of the feature vector that belongs to categories i and j, and c refers to the number of categories.

We use PCA to unify the dimensions of different features and calculate the interclass dispersion coefficient S_b for quantitative comparison. According to the value of the average dispersion value between classes S_b, it can be seen that, compared with the original features and the extracted features of the baseline method, DSR-GCN

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

TABLE XI
RUNNING TIME (S) AND PARAMETER SIZE OF DIFFERENT METHODS

Class	Time	RF [4]	2D-CNN [19]	SSRN [21]	SSUN [22]	DCFSL [40]	3DAES [57]	SSFTT [33]	ASPCNet [15]	CEGCN [61]	**DSR-GCN (Ours)**
Indian Pines	Train	0.5	9.6	9.3	26.1	3867.9	7	9.2	39.9	7.7	37.6
	Test	0.2	1.6	12.6	1.6	7	2.9	1.2	30	0.4	0.3
	Parameters	–	115,152	3637,687	1215,888	4270,121	269,584	218,504	3064,105	170,978	346,301
University of Pavia	Train	2.1	9.3	3.1	13.1	3948	3.9	4.5	56.6	87.9	148.5
	Test	4.1	6.9	73.1	1.9	51.8	11.6	5.2	251	4.3	2.9
	Parameters	–	111,401	216,312	1196,795	4259,294	138,128	218,049	1870,009	157,465	3817,421
Salinas	Train	2.1	11.1	10.71	29.8	536.2	6.9	10.3	53	24.4	232.7
	Test	3.6	8	63.8	14.9	36.7	16.2	9.5	116.9	2.3	1.7
	Parameters	–	115,152	148,728	1216,912	4270,521	269,584	218,504	2572,585	171,498	397,133
Houston	Train	6.6	10.4	3.9	7.7	1595	36.9	8.5	29.8	519.7	547.3
	Test	14.7	0.6	200.4	0.9	4.2	4	2.4	35.3	15.3	9.2
	Parameters	–	113,999	102,648	407,565	4264,360	751,823	218,439	2472,217	163,569	306,362

Note: "–" means that there are no parameters involved.

expands the dispersion of features between different classes.

B. Complexity Analysis

To analyze the computational complexity of DSR-GCN, we compare RF, 2-D CNN, SSRN, SSUN, DCFSL, 3DAES, SSFTT, ASPCNet, CEGCN, and our proposed DSR-GCN using five labeled samples per class in four datasets. The obtained training and test times, and the parameters' size are reported in Table XI. Specifically, we measure the number of network parameters by counting the number of them.

In terms of the network parameters, our network is composed of dual branches, in which S3Net and differentiated-scale GCN are integrated to fully absorb each other's advantage. Otherwise, in the GCN branch, the parameters approximately double because of our strategy of using a differentiated segmentation scale.

To avoid the overfitting problem, we adopt a lightweight depth separable convolution and GAP replacing part of the convolution step to reduce the parameters in the S3Net. In terms of training time, our method spends more training time for the larger dataset because the computation of graph convolution is very relevant to the size of the input dataset. Take a comprehensive consideration of the classification performance and the complexity of our method, the time cost and the total parameters are acceptable.

VI. CONCLUSION

In this article, we propose a novel DSR-GCN for few-shot HSI classification. In order to build a reliable initial graph, we propose to construct a graph with different scales, which considers both the subtle and relative wider range spatial spectral relation. In addition, we design an adaptive weighted feature fusion strategy and restricted fusion loss to balance the fusion of features extracted by differentiated-scale GCN branches, which helps to relieve the oversmoothing problem that might happen in the large-scale case. Nevertheless, a lightweight S3Net is designed integrated with differentiated-scale GCN to remedy the local pixel-level features, of which the learned features are distant for between classes and close for within class.

Although we have obtained encouraged experimental results, we will investigate to improve the proposed method. Currently, the differentiated-scale GCN branch only indirectly participates in the contrast learning process in the training of the S3Net branch, and we can further explore designing siamese GCN so that GCN can directly benefit from the contrast learning phase. Otherwise, we can introduce the ideas from active learning to produce more valuable training samples, which can help to obtain a more reliable graph in graph construction.

ACKNOWLEDGMENT

The authors would like to thank Prof. D. Landgrebe for making the Airborne Visible/Infrared Imaging Spectrometer Indian Pines hyperspectral dataset available to the community and Prof. P. Gamba for providing the Reflective Optics Spectrographic Imaging System data over Pavia, Italy.

REFERENCES

[1] D. Landgrebe, "Hyperspectral image data analysis," *IEEE Signal Process. Mag.*, vol. 19, no. 1, pp. 17–28, Jan. 2002.

[2] P. Ghamisi et al., "New frontiers in spectral–spatial hyperspectral image classification: The latest advances based on mathematical morphology, Markov random fields, segmentation, sparse representation, and deep learning," *IEEE Geosci. Remote Sens. Mag.*, vol. 6, no. 3, pp. 10–43, Sep. 2018.

[3] F. Melgani and L. Bruzzone, "Classification of hyperspectral remote sensing images with support vector machines," *IEEE Trans. Geosci. Remote Sens.*, vol. 42, no. 8, pp. 1778–1790, Aug. 2004.

[4] T. Kam Ho, "Random decision forests," in *Proc. 3rd Int. Conf. Document Anal. Recognit.*, vol. 1, Aug. 1995, pp. 278–282.

[5] Y. Y. Tang, H. Yuan, and L. Li, "Manifold-based sparse representation for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 52, no. 12, pp. 7606–7618, Dec. 2014.

[6] Y. Gu, T. Liu, X. Jia, and J. A. Benediktsson, "Nonlinear multiple kernel learning with multiple-structure-element extended morphological profiles for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 6, pp. 3235–3247, Jun. 2016.

[7] B. Zhang et al., "Progress and challenges in intelligent remote sensing satellite systems," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1814–1822, 2022.

[8] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[9] Y. Chen, Z. Lin, X. Zhao, G. Wang, and Y. Gu, "Deep learning-based classification of hyperspectral data," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 7, no. 6, pp. 2094–2107, Jun. 2014.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

[10] W. Hu, Y. Huang, L. Wei, F. Zhang, and H. Li, "Deep convolutional neural networks for hyperspectral image classification," *J. Sensors*, vol. 2015, Jul. 2015, Art. no. 258619.

[11] Y. Chen, X. Zhao, and X. Jia, "Spectral–spatial classification of hyperspectral data based on deep belief network," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 8, no. 6, pp. 2381–2392, Jun. 2015.

[12] L. Mou, P. Ghamisi, and X. X. Zhu, "Deep recurrent neural networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 55, no. 7, pp. 3639–3655, Jul. 2017.

[13] L. Zhu, Y. Chen, P. Ghamisi, and J. A. Benediktsson, "Generative adversarial networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 9, pp. 5046–5063, Sep. 2018.

[14] S. Sabour, N. Frosst, and G. E. Hinton, "Dynamic routing between capsules," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2017, pp. 3856–3866.

[15] J. Wang, X. Tan, J. Lai, and J. Li, "ASPCNet: Deep adaptive spatial pattern capsule network for hyperspectral image classification," *Neurocomputing*, vol. 486, pp. 47–60, May 2022.

[16] Q. Liu, F. Zhou, R. Hang, and X. Yuan, "Bidirectional-convolutional LSTM based spectral–spatial feature learning for hyperspectral image classification," *Remote Sens.*, vol. 9, no. 12, p. 1330, Dec. 2017.

[17] J. Yue, W. Zhao, S. Mao, and H. Liu, "Spectral–spatial classification of hyperspectral images using deep convolutional neural networks," *Remote Sens. Lett.*, vol. 6, no. 6, pp. 468–477, May 2015.

[18] J. Yue, S. Mao, and M. Li, "A deep learning framework for hyperspectral image classification using spatial pyramid pooling," *Remote Sens. Lett.*, vol. 7, no. 9, pp. 875–884, 2016.

[19] Y. Chen, H. Jiang, C. Li, X. Jia, and P. Ghamisi, "Deep feature extraction and classification of hyperspectral images based on convolutional neural networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 54, no. 10, pp. 6232–6251, Oct. 2016.

[20] S. Mei, J. Ji, Y. Geng, Z. Zhang, X. Li, and Q. Du, "Unsupervised spatial–spectral feature learning by 3D convolutional autoencoder for hyperspectral classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 9, pp. 6808–6820, Apr. 2019.

[21] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral–spatial residual network for hyperspectral image classification: A 3-D deep learning framework," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 2, pp. 847–858, Feb. 2018.

[22] Y. Xu, L. Zhang, B. Du, and F. Zhang, "Spectral–spatial unified networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 10, pp. 5893–5909, Oct. 2018.

[23] J. Zhu, L. Fang, and P. Ghamisi, "Deformable convolutional neural networks for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 15, no. 8, pp. 1254–1258, Aug. 2018.

[24] J. Nie, Q. Xu, J. Pan, and M. Guo, "Hyperspectral image classification based on multiscale spectral–spatial deformable network," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[25] Z. Gong, P. Zhong, Y. Yu, W. Hu, and S. Li, "A CNN with multiscale convolution and diversified metric for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 6, pp. 3599–3618, Jun. 2019.

[26] D. Wang, B. Du, L. Zhang, and Y. Xu, "Adaptive spectral–spatial multiscale contextual feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2461–2477, Mar. 2021.

[27] F. Cao and W. Guo, "Deep hybrid dilated residual networks for hyperspectral image classification," *Neurocomputing*, vol. 384, pp. 170–181, Apr. 2020.

[28] Y. Shen et al., "Efficient deep learning of nonlocal features for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 6029–6043, Jul. 2021.

[29] Y. Yuan, C. Wang, and Z. Jiang, "Proxy-based deep learning framework for spectral–spatial hyperspectral image classification: Efficient and robust," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–15, 2021.

[30] S. Li, X. Luo, Q. Wang, L. Li, and J. Yin, "H2AN: Hierarchical homogeneity-attention network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509816.

[31] Z. Xue, M. Zhang, Y. Liu, and P. Du, "Attention-based second-order pooling network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 11, pp. 9600–9615, Nov. 2021.

[32] J. He, L. Zhao, H. Yang, M. Zhang, and W. Li, "HSI-BERT: Hyperspectral image classification using the bidirectional encoder representation from transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 1, pp. 165–178, Jan. 2020.

[33] L. Sun et al., "Spectral–spatial feature tokenization transformer for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5522214.

[34] J. Feng et al., "Attention multibranch convolutional neural network for hyperspectral image classification based on adaptive region search," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 6, pp. 5054–5070, Jun. 2021.

[35] X. Liang, Y. Zhang, and J. Zhang, "Water retrieval embedded attention network with multiscale receptive fields for hyperspectral image refined classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5509022.

[36] X. He, Y. Chen, and P. Ghamisi, "Heterogeneous transfer learning for hyperspectral image classification based on convolutional neural network," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3246–3263, May 2020.

[37] R. Hang, Z. Li, Q. Liu, P. Ghamisi, and S. S. Bhattacharyya, "Hyperspectral image classification with attention-aided CNNs," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 3, pp. 2281–2293, Mar. 2021.

[38] S. J. Pan and Q. Yang, "A survey on transfer learning," *IEEE Trans. Knowl. Data Eng.*, vol. 22, no. 10, pp. 1345–1359, Oct. 2010.

[39] J. Yang et al., "Hyperspectral image classification using two-channel deep convolutional neural network," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Nov. 2016, pp. 5079–5082.

[40] Z. Li, M. Liu, Y. Chen, Y. Xu, W. Li, and Q. Du, "Deep cross-domain few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5501618.

[41] Y. Wang et al., "Heterogeneous few-shot learning for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[42] B. Settles, "Active learning literature survey," Univ. Wisconsin, Madison, WI, USA, Tech. Rep. 52, Jul. 2010.

[43] Y. Sun et al., "Active learning based autoencoder for hyperspectral imagery classification," in *Proc. IEEE Int. Geosci. Remote Sens. Symp.*, Nov. 2016, pp. 469–472.

[44] C. Deng et al., "Active transfer learning network: A unified deep joint spectral–spatial feature learning model for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 3, pp. 1741–1754, Mar. 2019.

[45] D. Ni and H. Ma, "Active learning for hyperspectral image classification using sparse code histogram and graph-based spatial refinement," *Int. J. Remote Sens.*, vol. 38, no. 3, pp. 923–948, Feb. 2017.

[46] B. Liu, X. Yu, A. Yu, P. Zhang, G. Wan, and R. Wang, "Deep few-shot learning for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 4, pp. 2290–2304, Apr. 2019.

[47] S. Jia, S. Jiang, Z. Lin, N. Li, M. Xu, and S. Yu, "A survey: Deep learning for hyperspectral image classification with few labeled samples," *Neurocomputing*, vol. 448, pp. 179–204, Aug. 2021.

[48] H. Tang et al., "A spatial–spectral prototypical network for hyperspectral remote sensing image," *IEEE Geosci. Remote Sens. Lett.*, vol. 17, no. 1, pp. 167–171, Jan. 2020.

[49] B. Xi, J. Li, Y. Li, R. Song, and Q. Du, "Deep prototypical networks with hybrid residual attention for hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 3683–3700, 2020.

[50] C. Zhang, J. Yue, and Q. Qin, "Global prototypical network for few-shot hyperspectral image classification," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 13, pp. 4748–4759, Aug. 2020.

[51] B. Deng and D. Shi, "Relation network for hyperspectral image classification," in *Proc. IEEE Int. Conf. Multimedia Expo Workshops (ICMEW)*, Jul. 2019, pp. 483–488.

[52] M. Rao, P. Tang, and Z. Zhang, "Spatial–spectral relation network for hyperspectral image classification with limited training samples," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 12, no. 12, pp. 5086–5100, Dec. 2019.

[53] A. Wang, C. Liu, D. Xue, H. Wu, Y. Zhang, and M. Liu, "Depthwise separable relation network for small sample hyperspectral image classification," *Symmetry*, vol. 13, no. 9, p. 1673, Sep. 2021.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.

[54] B. Liu, X. Yu, P. Zhang, A. Yu, Q. Fu, and X. Wei, "Supervised deep feature extraction for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 56, no. 4, pp. 1909–1921, Apr. 2018.

[55] Z. Li, X. Tang, W. Li, C. Wang, C. Liu, and J. He, "A two-stage deep domain adaptation method for hyperspectral image classification," *Remote Sens.*, vol. 12, no. 7, p. 1054, Mar. 2020.

[56] Z. Cao et al., "3D convolutional Siamese network for few-shot hyperspectral classification," *J. Appl. Remote Sens.*, vol. 14, no. 4, Nov. 2020, Art. no. 048504.

[57] S. Jia et al., "A semisupervised Siamese network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5516417.

[58] Z. Xue, Y. Zhou, and P. Du, "S3Net: Spectral–spatial Siamese network for few-shot hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5531219.

[59] A. Qin, Z. Shang, J. Tian, Y. Wang, T. Zhang, and Y. Y. Tang, "Spectral–spatial graph convolutional networks for semisupervised hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 16, no. 2, pp. 241–245, Sep. 2019.

[60] S. Wan, C. Gong, P. Zhong, B. Du, L. Zhang, and J. Yang, "Multiscale dynamic graph convolutional network for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 5, pp. 3162–3177, May 2020.

[61] Q. Liu, L. Xiao, J. Yang, and Z. Wei, "CNN-enhanced graph convolutional network with pixel-and superpixel-level feature fusion for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 10, pp. 8657–8671, Oct. 2021.

[62] D. Hong, L. Gao, J. Yao, B. Zhang, A. Plaza, and J. Chanussot, "Graph convolutional networks for hyperspectral image classification," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5966–5978, Jul. 2021.

[63] Y. Ding, Y. Chong, S. Pan, Y. Wang, and C. Nie, "Spatial–spectral unified adaptive probability graph convolutional networks for hyperspectral image classification," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Sep. 23, 2021.

[64] Y. Ding, Y. Guo, Y. Chong, S. Pan, and J. Feng, "Global consistent graph convolutional network for hyperspectral image classification," *IEEE Trans. Instrum. Meas.*, vol. 70, pp. 1–16, 2021.

[65] J. Yang et al., "Adaptive cross-attention-driven spatial–spectral graph convolutional network for hyperspectral image classification," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.

[66] R. Achanta et al., "SLIC superpixels compared to state-of-the-art superpixel methods," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 34, no. 11, pp. 2274–2282, Nov. 2012.

[67] J. Hu, L. Shen, S. Albanie, G. Sun, and E. Wu, "Squeeze-and-excitation networks," 2017, *arXiv:1709.01507*.

Zhaohui Xue (Member, IEEE) received the B.S. degree in geomatics engineering from Shandong Agricultural University, Tai'an, China, in 2009, the M.E. degree in remote sensing from the China University of Mining and Technology, Beijing, China, in 2012, and the Ph.D. degree in cartography and geographic information systems from Nanjing University, Nanjing, China, in 2015.

He is currently a Youth Professor (Ph.D. Supervisor) with the School of Earth Sciences and Engineering, Hohai University, Nanjing. He has authored more than 50 scientific papers, including more than 30 Science Citation Index (SCI) papers. His research interests include hyperspectral image classification, time-series image analysis, pattern recognition, and machine learning.

Dr. Xue is an Editorial Board Member of *National Remote Sensing Bulletin* for the term 2020–2024. He was a recipient of the National Scholarship for Doctoral Graduate Students granted by the Ministry of Education of the People's Republic of China in 2014. He was awarded the Best Reviewer for the IEEE Geoscience and Remote Sensing Society. He has been a Reviewer for more than ten famous remote sensing journals, including IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and *ISPRS Journal of Photogrammetry and Remote Sensing*.

Zhiwei Liu received the B.S. degree in geomatics engineering from Hohai University, Nanjing, China, in 2021, where he is currently pursuing the M.E. degree in surveying and mapping engineering.

His research interests include graph convolutional networks for hyperspectral image classification.

Mengxue Zhang received the B.S. and M.E. degrees in geomatics engineering from Hohai University, Nanjing, China, in 2018 and 2021, respectively. He is currently pursuing the Ph.D. degree in electrical engineering with the University of València, València, Spain.

His research interests include hyperspectral image classification and hybrid machine learning for drought and heatwave detection.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 07,2026 at 00:19:48 UTC from IEEE Xplore. Restrictions apply.